

# Endpoint crossings for random walks with memory

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## Abstract

Kagey asks when the middle bin of a Galton board is as likely as its ends if the ball tends to keep its direction. For the persistent random walk, which switches with probability  $T/n$  at each step, the crossing value  $z_n$  of  $T$  is about  $\log n$ . We ask what memory changes. In the elephant random walk each step copies a uniformly chosen earlier step and flips it with probability  $\varepsilon$ . We prove that the crossing  $\varepsilon_n$  is unique and that  $x_n = n\varepsilon_n$  satisfies  $x_n + \log x_n = 2 \log n - \log 8 + O((\log n)^2/n)$ . So  $x_n = 2z_n - \log(2\pi) + O(1/\log n)$ , about twice the Markov value. For large  $n$  the law at the crossing is not unimodal. Each end is below its neighbor, bin  $n - m$  has probability about  $x_n/(2m^2)$  for  $x_n \log^2 x_n \ll m \ll n$ , and the center is below every interior bin outside a window of width  $O(\sqrt{n} \log n)$ . So the profile has a hump near each end. For alternating runs with  $\mathbb{P}(\xi \geq k) = k^{-\alpha}$ ,  $1 < \alpha < 2$ , and a stationary start, the ratio of end to center is of order  $n^{1-\alpha+1/\alpha}$ . So the ends win for large  $n$  exactly when  $\alpha$  is below the golden ratio. A short remark treats aging walks, where the crossing with a fixed first step is exactly  $c = 1$ .

## 1 Introduction

Problem 131 of Kagey's Open Problem Collection [1] asks about a Galton board where the ball tends to keep going the same way, and in particular when the middle bin and the ends are equally likely. We count  $n$  steps, so the bins are Kagey's row  $n + 1$ , and we put  $L = \log n$ . Let  $A_n$  be the number of right steps. Unless we say otherwise, the first step is left or right with probability  $\frac{1}{2}$  each, so the law of  $A_n$  is symmetric about  $n/2$ . For even  $n$  put  $h = n/2$ ,  $C_n = \mathbb{P}(A_n = h)$  and  $E_n = \mathbb{P}(A_n = n) = \mathbb{P}(A_n = 0)$ . The crossing is the parameter value at which  $C_n = E_n$ .

The Markov walk switches direction with probability  $T/n$  at each step. This is the persistent random walk, and Paper A [2] brackets its crossing value  $z_n$  of  $T$  for every  $n \geq 2$ . From this bracket Paper A derives the crossing equation

$$z_n + \frac{1}{2} \log z_n = L + c_0 + \frac{1}{8z_n} + R_n, \quad c_0 = \frac{1}{2} \log \frac{\pi}{8}, \quad (1)$$

with an explicit remainder that satisfies  $|R_n| \leq \frac{1}{4L^2} + \frac{L^2+L}{n}$  for every  $n \geq 2$ . So  $z_n = L - \frac{1}{2} \log L + c_0 + o(1)$ , and a Markov walk needs about  $\log n$  switches before the middle catches up with the ends.

The Markov walk remembers only its last step. We compare it with 2 walks that have a longer memory. The elephant random walk of Schütz and Trimper [5] remembers its whole past. Each step copies a uniformly chosen earlier step and flips it with probability  $\varepsilon$ , and we put  $x = n\varepsilon$ . In the second walk the steps come in runs of alternating direction, and the runs after the first one have i.i.d. lengths  $\xi$  with  $\mathbb{P}(\xi \geq k) = k^{-\alpha}$ . Its memory is the time since the last switch. The first

run is either a fresh run or, for  $\alpha > 1$ , the remaining part of a run in progress. We call these the fresh and the stationary start. A remark treats a third kind of memory, aging, where the walk switches at step  $k$  with probability  $c/k$ .

**The elephant walk.** The ratio  $C_n/E_n$  is strictly increasing in  $\varepsilon$ , so the crossing  $\varepsilon_n$  exists and is unique (Proposition 2.3). The derivative of  $C_n$  at  $\varepsilon = 0$  is exactly  $4/(n+2)$  (Proposition 2.5), and  $\frac{4\varepsilon}{n+2}(1-2\varepsilon) \leq C_n \leq \frac{4\varepsilon}{n+2}n^{2\varepsilon}e^{5200\varepsilon}$  when  $n \geq 64$  and  $\varepsilon(1+\log n) \leq \frac{1}{100}$  (Theorem 2.6). Hence  $x_n = n\varepsilon_n$  satisfies  $x_n + \log x_n = 2\log n - \log 8 + O((\log n)^2/n)$  (Theorem 2.7). The factor 2 appears because a single early flip already turns a large family of steps. So the middle is reached with probability of order  $\varepsilon/n$ , and the ends must fall to about  $n^{-2}$  before they lose, instead of about  $n^{-1}$  for the Markov walk. As  $E_n \approx \frac{1}{2}e^{-x}$ , the crossing is near  $2\log n$ , not  $\log n$ . From (1) we get  $x_n = 2z_n - \log(2\pi) + c_*/\log n + O(\log \log n/(\log n)^2)$  with  $c_* = \frac{1}{2}\log(2\pi) - \frac{1}{4}$  (Theorem 2.8).

At the crossing each end is below its neighbor, so the law is not unimodal. We prove this for every even  $n \geq 1688$  (Theorem 3.5 and Corollary 3.6), and a computation gives it for every even  $8 \leq n < 1688$ . The number of steps that differ from the first step is close to a compound Poisson law, with a Landau limit near  $x \log x$  on the scale  $x$  (Remark 3.7, proofs omitted). Farther out,  $\mathbb{P}(A_n = n - m) \sim x/(2m^2)$  for  $x \log^2 x \ll m \ll n$  (Theorem 3.8). At the crossing and for large  $n$ , the center is below every interior bin outside a window of width of order  $(nx_n \log n)^{1/2}$ , that is  $n^{1/2} \log n$  (Theorem 3.9), and the modes lie within  $O(x_n \log^2 x_n)$  of the ends (Corollary 3.11). Conjecture 3.10 would close this window, and we checked it for every  $10 \leq n \leq 600$  and for 6 larger  $n$ .

**Heavy-tailed runs and aging.** For  $1 < \alpha < 2$  and any first run, Theorem 4.4 gives  $C_n$  to leading order, and it is of order  $n^{-1/\alpha}$ . With a stationary start the end is of order  $n^{1-\alpha}$ , so  $R_n = E_n/C_n \sim C(\alpha)n^{1-\alpha+1/\alpha}$  (Corollary 4.5). The end needs one first run that covers all  $n$  steps, while the center needs the renewal processes of  $+$  and  $-$  runs to meet, and these fluctuate on the scale  $n^{1/\alpha}$ . The exponent vanishes at the golden ratio  $\varphi = (1 + \sqrt{5})/2$ , and there  $R_n \rightarrow C(\varphi) = 0.7761\dots$ , so the center still wins. With a fresh start the center wins for every  $1 < \alpha < 2$ . Remark 4.6 lists the other exponents, with the proofs omitted (for  $\alpha = 1$  we only have a heuristic). For aging with a fixed first step the crossing is exactly  $c = 1$  for every  $n$  (Remark 4.8).

**Prior work.** The steps of the elephant walk split into clusters, since a step either copies an earlier step exactly, with probability  $1 - 2\varepsilon$ , or starts a new cluster with a fair sign. This is Kürsten's coupling with bond percolation on random recursive trees [6], and Baur and Bertoin [7] relate the walk to Pólya-type urns. Our regime  $n\varepsilon \asymp \log n$  lies in the strongly supercritical percolation regime of Baur [8] (see also Bertoin [9]). Those results describe cluster sizes and do not cover the center probability, which is a large deviation of the root cluster. Guérin, Laulin, Raschel and Simon [10, proof of Theorem 1.1] prove that the law with a fixed first step is unimodal for every  $n$  and every memory parameter, and their other results fix this parameter as  $n$  grows. We did not find an asymptotic formula for the center probability in the literature when  $\varepsilon \rightarrow 0$  as  $n$  grows.

For heavy-tailed runs, Godrèche and Luck [11] give the transform of the occupation time of an alternating renewal process and the scalings of the center and of both edges. Zaburdaev, Denisov and Klafter [13] and Dhar, Saito and Derrida [14] describe the profile of a Lévy walk. Wang, Li and Hänggi [15] already observe that equal decay of the central peak and the side peaks gives the golden ratio, as a heuristic for heat conduction. We prove the lattice statement with a local limit at the center, keep the 2 starts apart, and compute  $C(\alpha)$ . For aging, Dietz and Sethuraman [22]

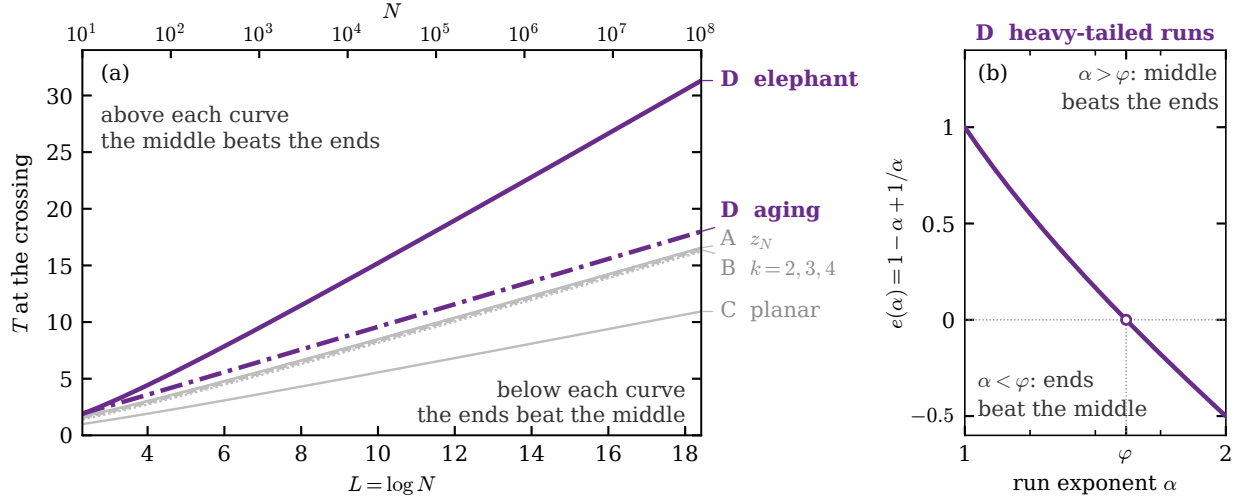


Figure 1: The crossing on each model's switching scale against  $L = \log N$  (a), and the heavy-tailed exponent  $e(\alpha) = 1 - \alpha + 1/\alpha$  of this paper for a stationary start (b). The scale is  $T$  for Papers A to C,  $x = n\varepsilon$  for the elephant walk (drawn through the root  $g_n$  of Theorem 2.7), and the expected number  $H_N - 1$  of switches at the fixed-start crossing  $c = 1$  for the aging walk. The curves of this paper are in bold.

proved the limit law of  $A_n/n$ , Engländer and Volkov [23] gave a second proof, and Engländer, Volkov and Wang [24] found the scaling limit of the walk.

**The 4 papers.** This is the last of 4 papers on Problem 131, and they share one notation. The walk takes  $N$  steps,  $L = \log N$ , and a Markov walk on  $d$  states has transition matrix  $P = I + (T/N)Q$ . We write  $z_N$  for the crossing value of  $T$  in Paper A [2]. That paper treats the 2-state walk, which is the zero-field Ising chain with free boundaries. It gives explicit uniform Bessel bounds for every bin, brackets the crossing  $p_N = 1 - z_N/N$  for every  $N \geq 2$  and derives the crossing equation (1). It also proves a uniform local law for every switching probability  $q \leq \frac{1}{2}$ , from the diffusive to the ballistic regime. Paper B [3] treats the complete graph, which is the  $d$ -state Potts chain. There every face (set of states) ties with the vertices at first order. Paper B resolves the tie at second order. Its constants  $a_k$  are a face's mass over a Gaussian volume, given by square roots of weighted spanning-tree sums, and the mode changes exactly once, from the vertices to the full centers. A start law or a weak field shifts the constants. Paper C [4] gives the general theory for  $P = I + (T/N)Q$  with  $T = \tau L$ . To first order the mode sits on a face  $F$  that minimizes  $\tau\lambda_F + |F| - 1$ , where  $\lambda_F$  is the principal Dirichlet eigenvalue of  $-Q$  on  $F$ . For strongly connected faces Paper C finds the second order, with constants  $K_F$  and a crossing equation, and it applies the theory to the planar persistent walk. This paper asks what memory changes. We write  $n$  for  $N$ , and  $x_n$  plays the role of  $z_n$ . The elephant walk crosses at  $x_n = 2z_n - \log(2\pi) + o(1)$ , about twice the Markov value, and its law at the crossing has a hump near each end. Heavy-tailed runs have the golden-ratio threshold, and aging walks get a short remark. As in Paper B the start matters. For heavy-tailed runs the threshold is  $\varphi$  for a stationary start (Corollary 4.5) and 1 for a fresh start (Remark 4.6, proofs omitted). Figure 1 shows the crossings of all 4 papers.

**Organization.** Sections 2 and 3 treat the elephant walk, Section 4 treats heavy-tailed runs and aging, Section 5 describes the computations, and Appendices A to C hold the longer proofs.

## 2 The elephant walk and its crossing

We reduce the elephant walk to a chain with a fixed first step, prove that the crossing is unique, and locate it. Throughout,  $n$  is even whenever  $C_n$  appears,  $h = n/2$  and  $L = \log n$ .

**Definition 2.1** (elephant walk). Fix  $0 < \varepsilon < 1$ . Let  $X_1$  be uniform on  $\{\pm 1\}$ . For  $t \geq 1$  let  $U_t$  be uniform on  $\{1, \dots, t\}$ , and put  $X_{t+1} = X_{U_t}$  with probability  $1 - \varepsilon$  and  $X_{t+1} = -X_{U_t}$  with probability  $\varepsilon$ . All these choices are independent. We call the event  $X_{t+1} = -X_{U_t}$  a flip. As before,  $A_n = \#\{i \leq n : X_i = +1\}$ . The crossing  $\varepsilon_n$  is the value of  $\varepsilon$  with  $C_n = E_n$ , and  $x_n = n\varepsilon_n$ .

In the notation of [5] the memory parameter is  $p = 1 - \varepsilon$ , and [10] write  $a = 2p - 1 = 1 - 2\varepsilon$ . We write  $\mathbb{P}_+$  and  $\mathbb{E}_+$  for the law and expectation given  $X_1 = +1$ . Let  $M_t$  be the number of  $i \leq t$  with  $X_i \neq X_1$ , and put  $r_t(b) = \varepsilon + (1 - 2\varepsilon)b/t$  and  $f_n(m) = \mathbb{P}_+(M_n = m)$ .

**Lemma 2.2.** *The process  $(M_t)_{t \geq 1}$  is a Markov chain with  $M_1 = 0$  and  $\mathbb{P}(M_{t+1} = b + 1 \mid M_t = b) = r_t(b) = 1 - \mathbb{P}(M_{t+1} = b \mid M_t = b)$ , and it is independent of  $X_1$ . For  $0 \leq k \leq n$ ,  $\mathbb{P}(A_n = k) = \frac{1}{2}[f_n(n - k) + f_n(k)]$ . In particular  $C_n = f_n(h)$ ,  $E_n = \frac{1}{2}(1 - \varepsilon)^{n-1}$ , and the law of  $A_n$  is symmetric about  $h$ .*

*Proof.* Given  $X_1, \dots, X_t$ , the copied step  $X_{U_t}$  differs from  $X_1$  with probability  $M_t/t$ . So  $X_{t+1} \neq X_1$  with conditional probability  $(1 - \varepsilon)M_t/t + \varepsilon(1 - M_t/t) = r_t(M_t)$ . This depends only on  $M_t$  and not on  $X_1$ , which proves the first claim. On  $\{X_1 = +1\}$  we have  $A_n = n - M_n$ , and on  $\{X_1 = -1\}$  we have  $A_n = M_n$ . Averaging over  $X_1$  gives the formula for  $\mathbb{P}(A_n = k)$ , which does not change under  $k \mapsto n - k$ . Finally  $M_n \leq n - 1$ , so  $f_n(n) = 0$ , and  $f_n(0) = \prod_{t=1}^{n-1} (1 - r_t(0)) = (1 - \varepsilon)^{n-1}$ .  $\square$

**Proposition 2.3.** *Let  $n \geq 2$  and  $1 \leq k \leq n - 1$ . Then  $f_n(k)/f_n(0)$  and  $\mathbb{P}(A_n = k)/E_n$  are strictly increasing in  $\varepsilon \in (0, 1)$ . In particular  $C_n/E_n$  is strictly increasing, the crossing  $\varepsilon_n$  exists and is unique, and  $\varepsilon_n < \frac{1}{2}$ . Also  $C_n \geq 2E_n$  for  $\varepsilon \geq \frac{1}{2}$ .*

*Proof.* The probability of a path of  $M$  is a product of  $n - 1$  factors, namely  $r_s(b_s)$  for an up step at time  $s$  and  $1 - r_s(b_s)$  otherwise, where  $b_s$  is the state at time  $s$ . We divide it by the probability  $f_n(0) = (1 - \varepsilon)^{n-1}$  of the path that never flips. With  $y = b_s/s \in [0, 1)$  the factors become  $\frac{\varepsilon + (1 - 2\varepsilon)y}{1 - \varepsilon}$  and  $1 - y \frac{1 - 2\varepsilon}{1 - \varepsilon}$ , with derivatives  $\frac{1 - y}{(1 - \varepsilon)^2} > 0$  and  $\frac{y}{(1 - \varepsilon)^2} \geq 0$  in  $\varepsilon$ . Both factors are positive, since  $r_s(b_s) = \varepsilon(1 - y) + (1 - \varepsilon)y$  lies strictly between 0 and 1. So the ratio for each path is nondecreasing in  $\varepsilon$ , and strictly increasing if the path has an up step. Summing over the paths with  $k \geq 1$  up steps proves the first claim, and the second follows by Lemma 2.2. As  $\varepsilon \rightarrow 0$ ,  $C_n \rightarrow 0$  and  $E_n \rightarrow \frac{1}{2}$ . For  $\varepsilon \geq \frac{1}{2}$  every  $r_s(b)$  lies in  $[1 - \varepsilon, \varepsilon]$ , so each path of  $M$  that ends at  $h$  has probability at least  $(1 - \varepsilon)^{n-1} = 2E_n$ . Hence  $C_n \geq 2E_n$ . The ratio  $C_n/E_n$  is continuous and strictly increasing, so it equals 1 at exactly one  $\varepsilon$ , and this  $\varepsilon$  lies in  $(0, \frac{1}{2})$ .  $\square$

After the first flip, and to first order in  $\varepsilon$ , the chain  $M$  is Pólya's urn.

**Lemma 2.4** (Pólya's urn). *Let  $2 \leq v \leq n$ . At time  $v$  an urn holds 1 ball of one color and  $v - 1$  of the other. At each time  $k = v, \dots, n - 1$  we draw a ball uniformly and return it with 1 more ball of its color. Let  $\beta_v(d)$  be the probability that at time  $n$  there are  $d$  balls of the first color. Then  $\beta_v(d) = \binom{n-d-1}{v-2} / \binom{n-1}{v-1}$  for  $1 \leq d \leq n - v + 1$ , and  $\beta_v(d) = 0$  for other  $d$ . Also  $\beta_v$  is nonincreasing in  $d$  and  $\sum_d d \beta_v(d) = n/v$ . For  $1 \leq d \leq n - 1$ ,*

$$\sum_{v=2}^n \beta_v(d) = \frac{n}{d(d+1)}, \quad \sum_{v=2}^n (v-2) \beta_v(d) = \frac{2n(n-d-1)}{d(d+1)(d+2)}.$$

*Proof.* If the urn holds  $s$  balls of the first color at time  $k$ , the next draw is of that color with probability  $s/k$ . So a given order of  $i$  draws of the first color among the  $n-v$  draws has probability  $\frac{i!(v-1)v\cdots(n-i-2)}{v(v+1)\cdots(n-1)} = \frac{(v-1)! (n-i-2)!}{(n-1)!}$ . Multiplying by  $\binom{n-v}{i}$  and putting  $d = i+1$  gives  $\beta_v(d) = a_d(v-1)\binom{n-v}{d-1}$  with  $a_d = \frac{(d-1)!(n-d-1)!}{(n-1)!}$ , which equals  $\binom{n-d-1}{v-2}/\binom{n-1}{v-1}$ . Hence  $\beta_v(d)/\beta_v(d-1) = 1 - \frac{v-2}{n-d}$  for  $2 \leq d \leq n-v+1$ , and  $\beta_v$  is nonincreasing. The fraction of balls of the first color is a martingale that starts at  $1/v$ , which gives the mean. For the 2 sums put  $i = n-v$ , so that  $v-1 = \binom{n-1-i}{1}$  and  $(v-1)(v-2) = 2\binom{n-1-i}{2}$ . By the Chu–Vandermonde identity,  $\sum_i \binom{i}{d-1} \binom{n-1-i}{1} = \binom{n}{d+1}$  and  $\sum_i \binom{i}{d-1} \binom{n-1-i}{2} = \binom{n}{d+2}$ . Multiplying by  $a_d$  and  $2a_d$  gives the 2 sums.  $\square$

Let  $V_s(b) = \mathbb{P}_+(M_n = h \mid M_s = b)$ . For  $1 \leq t \leq h$  let  $\pi_t = \beta_{t+1}(h) = \frac{t}{n-t} \prod_{i=1}^{t-1} \frac{h-i}{n-i}$ ,  $c_1 = \sum_{t=1}^h \pi_t$  and  $w_t = \pi_t/c_1$ , so that  $(w_t)_{1 \leq t \leq h}$  is a probability vector.

**Proposition 2.5.** *Let  $n \geq 2$  be even.*

(a)  $C_n = \sum_{t=1}^h \varepsilon(1-\varepsilon)^{t-1} V_{t+1}(1)$ .

(b)  $C_n$  is a polynomial in  $\varepsilon$  with  $C_n(0) = 0$ . At  $\varepsilon = 0$  we have  $V_{t+1}(1) = \pi_t$ , and  $C'_n(0) = c_1 = \frac{4}{n+2}$  exactly.

(c)  $\sum_t w_t t = 3 - \frac{6}{h+2}$ , and  $w_t \leq 2t2^{-t}$  for  $1 \leq t \leq h$ .

*Proof.* (a) Under  $\mathbb{P}_+$  the first flip happens at step  $t+1$  exactly when  $M_s = 0$  for  $s \leq t$  and  $M_{t+1} = 1$ . Since  $r_s(0) = \varepsilon$ , this has probability  $(1-\varepsilon)^{t-1}\varepsilon$ , and by the Markov property the chain then ends at  $h$  with probability  $V_{t+1}(1)$ . At time  $t+1$  there are  $t$  steps equal to  $X_1$ , and this number never decreases. So  $V_{t+1}(1) = 0$  for  $t > h$ . Paths with no flip end at  $0 \neq h$ .

(b) Every transition probability is affine in  $\varepsilon$ , so  $C_n$  and  $V_{t+1}(1)$  are polynomials in  $\varepsilon$ . By (a),  $C_n(0) = 0$  and  $C'_n(0) = \sum_t V_{t+1}(1)|_{\varepsilon=0}$ . At  $\varepsilon = 0$  we have  $r_s(b) = b/s$ . So from time  $t+1$  on,  $M$  counts the balls of the first color in the urn of Lemma 2.4 with  $v = t+1$ , and  $V_{t+1}(1) = \beta_{t+1}(h) = \pi_t$ . Since  $\beta_v(h) = 0$  for  $v > h+1$ , Lemma 2.4 with  $d = h$  gives  $c_1 = \sum_{v=2}^n \beta_v(h) = \frac{n}{h(h+1)} = \frac{4}{n+2}$ .

(c) In the same way the second sum of Lemma 2.4 gives  $\sum_t (t-1)\pi_t = \frac{2n(h-1)}{h(h+1)(h+2)}$ , so  $\sum_t w_t(t-1) = \frac{2(h-1)}{h+2}$ . Next  $\frac{h-i}{n-i} \leq \frac{1}{2}$ , so  $w_t = \frac{n+2}{2(n-t)} \pi_t \leq \frac{n+2}{2(n-t)} t2^{-t} \leq 2t2^{-t}$  for  $t \leq h$ .  $\square$

So to first order the walk flips once, at some step  $t+1$ , and then runs as Pólya's urn from  $t$  balls of one color and 1 of the other. For large  $n$  the minority fraction is close to a Beta(1,  $t$ ) variable, whose density at  $\frac{1}{2}$  is  $t2^{1-t}$ , and  $\sum_t t2^{1-t} = 4$  explains the constant  $4/(n+2)$ .

**Theorem 2.6.** *Let  $n \geq 2$  be even.*

(a) For  $0 < \varepsilon < \frac{1}{2}$ ,  $C_n \geq \frac{4\varepsilon}{n+2}(1-2\varepsilon)$ .

(b)  $C_n \leq \frac{4\varepsilon}{n+2} n^{2\varepsilon} e^{5200\varepsilon}$  whenever  $n \geq 64$  and  $0 < \varepsilon(1+\log n) \leq \frac{1}{100}$ .

In particular  $\log C_n = \log \frac{4\varepsilon}{n+2} + O(\varepsilon \log n)$  in the range of (b).

After the first flip,  $M$  at time  $s$  is Pólya's urn with  $\beta s$  extra balls of each color, where  $\beta = \varepsilon/(1-2\varepsilon)$  (Lemma A.1). They pull the minority fraction toward  $\frac{1}{2}$ , which adds about  $2\varepsilon/s$  to the log-probability of a path to the center at time  $s$ , and summing over  $s \leq n$  gives  $n^{2\varepsilon}$ . The lower bound (a) compares  $M$  with Pólya's urn itself, and the upper bound (b) with an urn whose pseudo-count  $\kappa(s+10)$  grows linearly, where  $\kappa = 2\varepsilon/(1-4\varepsilon)$  (Appendix A).

We now locate the crossing. By Theorem 2.6,  $C_n \approx 4x/n^2$  when  $\varepsilon = x/n$  and  $x$  is of order  $\log n$ . Also  $E_n = \frac{1}{2}(1-x/n)^{n-1} \approx \frac{1}{2}e^{-x}$ . Setting them equal gives  $x + \log x \approx 2\log n - \log 8$ . So let  $\Lambda_n = 2L - \log 8$  and  $G(x) = x + \log x$ , and let  $g_n$  be the root of  $G(g_n) = \Lambda_n$ .

| $n$   | $x_n$     | $g_n$     | $x_n - x_{2\text{nd}}$ | $x_n / \log n$ | $D(n) \log n$ |
|-------|-----------|-----------|------------------------|----------------|---------------|
| 50    | 3.940487  | 4.288636  | +0.3452                | 1.007          | -0.083        |
| 400   | 7.629217  | 7.843768  | +0.0621                | 1.273          | -0.199        |
| 1600  | 10.240565 | 10.340051 | +0.0179                | 1.388          | +0.215        |
| 3200  | 11.546834 | 11.610464 | +0.0097                | 1.431          | +0.386        |
| 12800 | 14.158845 | 14.182921 | +0.0028                | 1.497          | +0.607        |
| 51200 | 16.778468 | 16.786946 | +0.0008                | 1.547          | +0.707        |

Table 1: Exact elephant crossings, the root  $g_n$  of  $G(g) = \Lambda_n$ , and the difference  $D(n)$  of Theorem 2.8 times  $\log n$ . Here  $x_{2\text{nd}}$  is the second-order formula of Remark 2.9 without its error term.

**Theorem 2.7.** *As  $n \rightarrow \infty$  over even  $n$ , the following hold.*

- (a)  $x_n + \log x_n = \Lambda_n + O((\log n)^2/n)$ , so  $x_n = g_n + O((\log n)^2/n)$ .
- (b)  $x_n = 2 \log n - \log \log n - \log 16 + O(\log \log n / \log n)$ .

*Proof.* (a) Put  $F(x) = \log C_n(x/n) - \log E_n(x/n)$  for  $0 < x < n$ . By Proposition 2.3,  $F$  is strictly increasing and  $x_n$  is its only zero. Let  $1 \leq x \leq 2\Lambda_n$ . Then  $x/n \leq 4L/n$ , so for large  $n$  Theorem 2.6 applies with  $\varepsilon = x/n$  and gives  $\log C_n = \log x + \log 4 - 2L + O(xL/n)$ . Also  $\log(1-y) = -y + O(y^2)$  for  $0 \leq y \leq \frac{1}{2}$ , so  $\log E_n = -\log 2 - x + O(x^2/n)$ . Since  $\log 4 + \log 2 = \log 8$ , we get  $|F(x) - G(x) + \Lambda_n| \leq \rho_n$  on  $[1, 2\Lambda_n]$ , with  $\rho_n = O(L^2/n)$ . Now  $G' \geq 1$  and  $G(g_n) = \Lambda_n$ . So  $F(x) < 0$  for  $1 \leq x < g_n - \rho_n$  and  $F(x) > 0$  for  $g_n + \rho_n < x \leq 2\Lambda_n$ . As  $\Lambda_n - \log \Lambda_n \leq g_n \leq \Lambda_n$ , for large  $n$  both ranges are nonempty. Hence  $|x_n - g_n| \leq \rho_n$ , and  $F(x_n) = 0$  gives  $|G(x_n) - \Lambda_n| \leq \rho_n$ .

(b) Since  $g_n = \Lambda_n - \log g_n$  and  $\Lambda_n - \log \Lambda_n \leq g_n \leq \Lambda_n$ , we have  $\log g_n = \log \Lambda_n + O(\log \Lambda_n / \Lambda_n)$ , and  $\log \Lambda_n = \log 2 + \log \log n + O(1/\log n)$ . This gives (b) for  $g_n$ , and so for  $x_n$  by (a).  $\square$

Table 1 gives exact crossings on the grid  $n = 50 \cdot 2^k$ , computed from the recursion for  $f_n$  in Lemma 2.2. The limit 2 of  $x_n / \log n$  is approached slowly.

Now we compare with the Markov walk. Doubling the crossing equation (1) gives  $2z_n + \log(2z_n) \approx \Lambda_n + \log(2\pi)$ .

**Theorem 2.8.** *Let  $z_n$  be the crossing value of  $T$  for the persistent random walk of [2] with  $n$  steps. As  $n \rightarrow \infty$  over even  $n$ ,*

$$D(n) := x_n - (2z_n - \log(2\pi)) = \frac{c_*}{\log n} + O\left(\frac{\log \log n}{(\log n)^2}\right), \quad c_* = \frac{1}{2} \log(2\pi) - \frac{1}{4} = 0.668939 \dots$$

*Proof.* Write  $\ell = \log(2\pi)$ . The crossing equation (1) of Paper A [2] holds with  $|R_n| \leq \frac{1}{4L^2} + \frac{L^2+L}{n}$  for every  $n \geq 2$ , and Paper A also shows  $L/2 \leq z_n \leq L$ . So  $R_n = O(L^{-2})$  and  $z_n = L - \frac{1}{2} \log L + O(1)$ . Since  $z_n \geq L/2$ , this gives  $\frac{1}{8z_n} - \frac{1}{8L} = \frac{L-z_n}{8Lz_n} = O(\log L/L^2)$ . Put  $Y = 2z_n$ . Since  $2c_0 + \log 2 = \log \frac{\pi}{4}$ , we get  $Y + \log Y = 2L + \log \frac{\pi}{4} + \frac{1}{4L} + O(\log L/L^2)$ . Also  $Y = 2L - \log L + O(1)$ , so  $\log(1 - \ell/Y) = -\ell/(2L) + O(\log L/L^2)$ . Using  $\log \frac{\pi}{4} - \ell = -\log 8$ , we get

$$G(Y - \ell) = Y + \log Y - \ell + \log\left(1 - \frac{\ell}{Y}\right) = \Lambda_n + \frac{1}{4L} - \frac{\ell}{2L} + O\left(\frac{\log L}{L^2}\right) = \Lambda_n - \frac{c_*}{L} + O\left(\frac{\log L}{L^2}\right).$$

By Theorem 2.7(a),  $G(x_n) = \Lambda_n + O(L^2/n)$ . Both  $x_n$  and  $Y - \ell = 2z_n - \log(2\pi)$  are  $2L + O(\log L)$ , so  $G' = 1 + O(1/L)$  between them, and the mean value theorem gives  $x_n - (Y - \ell) = c_*/L + O(\log L/L^2)$ .  $\square$

We computed  $z_n$  from the exact law of the Markov walk [2]. The difference  $D(n)$  changes sign between  $n = 400$  and  $n = 800$ , and  $D(n) \log n$  is 0.707 at  $n = 51200$  against the limit  $c_* = 0.669$ . So the asymptotic regime is reached slowly.

*Remark 2.9* (second order, proofs omitted). Write  $V_{t+1}(1)/\pi_t$  as the mean of a product of factors  $1 + O(\varepsilon)$  along Pólya's urn conditioned to end at  $h$ . Jensen's inequality then gives a lower bound that matches Theorem 2.6(b), so  $\log C_n = \log \frac{4\varepsilon}{n+2} + 2\varepsilon \log n + O(\varepsilon)$  in the range of that theorem. With this the proof of Theorem 2.7 gives  $x_n = g_n - \frac{2g_n \log n + g_n^2/2}{n(1+1/g_n)} + O(\frac{\log n}{n})$ , so  $x_n < g_n$  for large  $n$ . Write  $C_n = c_1\varepsilon + c_2\varepsilon^2 + O(\varepsilon^3)$  and  $k_2(n) = c_2/c_1$ . A longer computation shows that  $k_2(n) - 2 \log n \rightarrow 2\gamma - 2 - \log 2 = -1.5387\dots$ , where  $\gamma$  is Euler's constant. We omit the details of these 3 results. We do not know an upper bound  $\log C_n \leq \log(c_1\varepsilon) + \varepsilon k_2(n) + O(\varepsilon^2(\log n)^k)$ , or an explicit  $n$  beyond which  $g_n > x_n$ .

### 3 The elephant walk near its crossing

We now look at the whole law of  $A_n$  near the crossing (Figure 2). Unless stated otherwise,  $n \geq 64$ ,  $0 < \varepsilon \leq \frac{1}{4}$  and  $x = n\varepsilon \geq 1$ . Recall that  $\mathbb{P}(A_n = n - m) = \frac{1}{2}[f_n(m) + f_n(n - m)]$ . Some numerical statements also use odd  $n$ , and there  $x_n$  is the value of  $x$  at which the 2 central bins are as likely as the ends (it is unique by Proposition 2.3). For a mode  $a^* \geq n/2$  of the law of  $A_n$  we put  $m^* = n - a^*$ , and  $m_h$  is the largest mode of  $f_n$ .

#### 3.1 Families and the first flip

For  $k_0 \geq 1$  let  $(M_k^{[k_0]})_{k \geq k_0}$  be the Markov chain with  $M_{k_0}^{[k_0]} = 0$  and the transition probabilities  $r_k(b)$  of  $M$ . So  $M^{[1]}$  has the law of  $M$ . We write  $Y_d = M_d^{[1]}$ , and  $Z_t^{(j)} = M_t^{[j-1]}$  for  $j \geq 2$  and  $t \geq j - 1$ . In a random recursive tree on  $[n] = \{1, \dots, n\}$ , vertex  $k + 1$  is joined to a uniform vertex of  $[k]$ , independently for  $k = 1, \dots, n - 1$ . We write  $\mathcal{T}_v$  for the subtree of  $v$ , that is,  $v$  and its descendants, and we call it the family of  $v$ . A single flip at  $v$  turns every step of its family.

**Lemma 3.1** (tree and marks). *Let  $\mathcal{T}$  be the tree on  $[n]$  in which the parent of  $k + 1$  is  $U_k$ . For  $2 \leq v \leq n$  let  $\mu_v = 1$  if step  $v$  flips the step it copies, and  $\mu_v = 0$  otherwise. Then  $\mathcal{T}$  is a random recursive tree, the marks  $\mu_v$  are i.i.d. Bernoulli( $\varepsilon$ ) and independent of  $\mathcal{T}$ , and  $X_v = X_1(-1)^{\sigma(v)}$ , where  $\sigma(v)$  is the number of marks on the path from  $v$  to the root, the root excluded. So  $M_n = \#\{v : \sigma(v) \text{ odd}\}$ . If the marks on  $2, \dots, k_0$  are set to 0, then the number of odd vertices in  $[k]$ ,  $k \geq k_0$ , is a copy of the chain  $M^{[k_0]}$ .*

*Proof.* The parent  $U_k$  of  $k + 1$  is uniform on  $[k]$  and independent of the earlier choices, and the marks are the flip indicators. The path from  $k + 1$  to the root is  $k + 1$  followed by the path from  $U_k$ , so  $\sigma(k + 1) = \mu_{k+1} + \sigma(U_k)$  and, by induction on  $k$ ,  $X_{k+1} = X_{U_k}(-1)^{\mu_{k+1}} = X_1(-1)^{\sigma(k+1)}$ . Now set  $\mu_2 = \dots = \mu_{k_0} = 0$ . If  $b$  of the vertices in  $[k]$  are odd,  $k \geq k_0$ , then  $k + 1$  is odd with conditional probability  $(1 - \varepsilon)\frac{b}{k} + \varepsilon(1 - \frac{b}{k}) = r_k(b)$ , whatever else is known about  $[k]$ .  $\square$

The coupling with a random recursive tree is due to Kürsten [6], who uses bond percolation with retention probability  $1 - 2\varepsilon$ . The marks above give the same law of the walk. For  $k \geq v$ , if  $\mathcal{T}_v$  contains  $s$  of the vertices in  $[k]$ , then  $k + 1$  joins  $\mathcal{T}_v$  with probability  $s/k$ . This is the urn of Lemma 2.4, so  $|\mathcal{T}_v|$  has the law  $\beta_v$  and mean  $n/v$ , and on average there are  $n/(d(d + 1))$  families of size  $d$ . If  $v$  is the only marked vertex, the odd vertices are those of  $\mathcal{T}_v$ . Two or more marks have probability  $O(\varepsilon^2)$ , and no mark gives  $M_n = 0$ . So at fixed  $n$  the part of  $f_n(m)$  of first order in  $\varepsilon$  is

exactly  $\varepsilon \sum_v \beta_v(m) = x/(m(m+1))$ , for  $1 \leq m \leq n-1$  (at  $m = h$  this is  $c_1 \varepsilon$  of Proposition 2.5). This is also proved in Lean (Section 5).

**Lemma 3.2** (splitting). *Let  $\mathcal{T}$  be a random recursive tree on  $[n]$ , let  $2 \leq v \leq n$  and let  $S \subseteq [n]$  with  $\min S = v$ . Given that  $\mathcal{T}_v$  has vertex set  $S$ , the restrictions of  $\mathcal{T}$  to  $S$  and to  $[n] \setminus S$ , relabeled in increasing order, are independent random recursive trees. They are independent of the parent of  $v$ , which is uniform on  $[v-1]$ .*

*Proof.* The tree  $\mathcal{T}$  is uniform over the  $(n-1)!$  increasing trees on  $[n]$ . Its family  $\mathcal{T}_v$  has vertex set  $S$  exactly when every  $u \in S \setminus \{v\}$  has its parent in  $S$  and every  $u \notin S$  with  $u \geq 2$  has its parent outside  $S$ . So, given this event, each element of  $S$  other than  $v$  picks its parent freely among the smaller elements of  $S$ ,  $v$  picks any element of  $[v-1]$ , and each element of  $[n] \setminus S$  other than 1 picks freely among the smaller elements of  $[n] \setminus S$ . This is the claim.  $\square$

**Lemma 3.3** (the first flip). *Let  $V = \min\{v \geq 2 : \mu_v = 1\}$ , with  $V = \infty$  if there is no mark, and let  $D = |\mathcal{T}_V|$ . Then  $\mathbb{P}(V = j, D = d) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$ , and for  $m \geq 1$*

$$f_n(m) = \sum_{j=2}^n \varepsilon(1-\varepsilon)^{j-2} \sum_{d=1}^{n-j+1} \beta_j(d) \mathbb{P}(d - Y_d + Z_{n-d}^{(j)} = m),$$

where  $Y_d$  and  $Z_{n-d}^{(j)}$  are independent.

*Proof.* Marks and tree are independent, which gives the law of  $(V, D)$ . Condition on  $V = j$  and on the vertex set  $S$  of  $\mathcal{T}_j$ , with  $|S| = d$ . The marks on  $j+1, \dots, n$  are still i.i.d., and by Lemma 3.2 the trees on  $S$  and on  $[n] \setminus S$  are independent random recursive trees. The vertices  $1, \dots, j-1$  carry no mark, so  $j$  is odd, and  $u \in S$  is odd exactly when the path from  $u$  up to  $j$ , with  $j$  excluded, has an even number of marks. So by Lemma 3.1 for the tree on  $S$  rooted at  $j$ , the number of odd vertices in  $S$  has the law of  $d - Y_d$ . Paths from outside  $S$  stay outside  $S$ , and the tree on  $[n] \setminus S$  has the unmarked vertices  $1, \dots, j-1$  first. So by Lemma 3.1 its odd count has the law of  $Z_{n-d}^{(j)}$ . The 2 counts are independent, and their law depends on  $S$  only through  $d$ . Sum over  $j$  and  $d$ , and note that  $M_n = 0$  if there is no mark.  $\square$

So the first flip turns its whole family except the steps of that family that flip back, and the rest of the tree is a walk whose first  $j-1$  steps agree with  $X_1$ . Only the law of each pair  $(Y_d, Z_{n-d}^{(j)})$  enters the lemma. So from now on  $(Y_d)_{d \geq 1}$  is one path of  $M^{[1]}$  and  $(Z_t^{(j)})_{t \geq j-1}$  one path of  $M^{[j-1]}$ , independent of each other and of  $(V, D)$ .

**Lemma 3.4** (unimodality given the first step). *The function  $f_n$  is unimodal.*

This is shown in the proof of Theorem 1.1 of Guérin, Laulin, Raschel and Simon [10], for every  $n$  and every memory parameter  $p = 1 - \varepsilon$ , and it is also proved in Lean (Section 5). So  $f_n$  is nondecreasing on  $\{0, \dots, m_h\}$  and nonincreasing on  $\{m_h, \dots, n\}$ .

### 3.2 The end and its neighbor

Bin  $n-1$  needs one flip that is never copied, and a flip at step  $j$  is never copied with probability about  $j/n$ . Summing  $\varepsilon j/n$  over  $j$  gives about  $x/2$ , so the end should fall below its neighbor once  $x > 2$ .

**Theorem 3.5** (endpoint dip). *Let  $n \geq 3$  and  $0 < \varepsilon \leq \frac{1}{2}$ , and put  $\rho_1 = \mathbb{P}(A_n = n - 1)/E_n$ . Then*

$$\frac{x}{2(1-\varepsilon)} < \rho_1 \leq \frac{x(1+1/n)^2}{2-3\varepsilon} + \frac{\varepsilon}{1-\varepsilon}.$$

*Hence the end is strictly below its neighbor if  $x \geq 2(1-\varepsilon)$ , and strictly above it if the right side is below 1.*

*Proof.* By Lemma 2.2,  $\rho_1 = (f_n(1) + f_n(n-1))/f_n(0)$  with  $f_n(0) = (1-\varepsilon)^{n-1}$ . On the event  $M_n = 1$  the chain first moves up from time  $j-1$  to time  $j$ , for some  $2 \leq j \leq n$ , and then stays at 1. Since  $r_t(0) = \varepsilon$  and  $1 - r_k(1) = (1-\varepsilon)(1-c/k)$  with  $c = (1-2\varepsilon)/(1-\varepsilon) \in [0, 1]$ ,

$$f_n(1) = \varepsilon(1-\varepsilon)^{n-2} \sum_{j=2}^n P_j, \quad P_j = \prod_{k=j}^{n-1} \left(1 - \frac{c}{k}\right).$$

Since  $c \leq 1$ ,  $P_j \geq \prod_{k=j}^{n-1} (1 - 1/k) = (j-1)/(n-1)$ , and these lower bounds sum to  $n/2$ . So  $f_n(1)/f_n(0) \geq x/(2(1-\varepsilon))$ . Next,  $1 - c/k \leq e^{-c/k}$  and  $\sum_{k=j}^{n-1} 1/k \geq \log(n/j)$  give  $P_j \leq (j/n)^c$ . As  $(t/n)^c$  is nondecreasing in  $t$ ,

$$\sum_{j=2}^n P_j \leq \int_2^{n+1} \left(\frac{t}{n}\right)^c dt \leq \frac{n(1+1/n)^{c+1}}{c+1} \leq \frac{n(1+1/n)^2(1-\varepsilon)}{2-3\varepsilon},$$

using  $c \leq 1$  and  $1/(c+1) = (1-\varepsilon)/(2-3\varepsilon)$ . So  $f_n(1)/f_n(0) \leq x(1+1/n)^2/(2-3\varepsilon)$ . Finally,  $M_n = n-1$  needs an up move at every time. The first has probability  $r_1(0) = \varepsilon$ , and the one at time  $k \geq 2$  has probability  $r_k(k-1) = 1-\varepsilon-(1-2\varepsilon)/k \in [\frac{1}{2}, 1-\varepsilon]$ . So  $0 < f_n(n-1) \leq \varepsilon(1-\varepsilon)^{n-2}$ . This makes the lower bound strict and gives the term  $\varepsilon/(1-\varepsilon)$ .  $\square$

**Corollary 3.6** (the dip at the crossing). *Let  $n \geq 4$  be even and suppose that  $x_n \geq 2(1-\varepsilon_n)$ . Then at the crossing  $E_n = C_n < \mathbb{P}(A_n = n-1)$ , and the law of  $A_n$  is not unimodal.*

*Proof.* By Proposition 2.3,  $\varepsilon_n < \frac{1}{2}$ , so Theorem 3.5 gives  $\mathbb{P}(A_n = n-1) > E_n = C_n$ . Suppose the law is nondecreasing on  $\{0, \dots, a_0\}$  and nonincreasing on  $\{a_0, \dots, n\}$ . It is symmetric about  $h$ , so we may replace  $a_0$  by  $n - a_0$  and assume  $a_0 \leq h$ . Then  $\mathbb{P}(A_n = a) \leq C_n$  for every  $a \geq h$ , including  $a = n-1$ , which is a contradiction.  $\square$

The hypothesis  $x_n \geq 2(1-\varepsilon_n)$  holds for every even  $n \geq 1688$ . To see this, take  $\varepsilon = 2/n$ . Then  $\varepsilon(1 + \log n) \leq \frac{1}{100}$ , so Theorem 2.6(b) gives  $C_n \leq \frac{8}{n(n+2)} n^{4/n} e^{10400/n}$ . The right side decreases in  $n$  and is below  $1.4 \cdot 10^{-3}$  at  $n = 1688$ . Since  $\log(1-y) \geq -y - y^2$  for  $0 \leq y \leq \frac{1}{2}$ , we also have  $E_n = \frac{1}{2}(1 - \frac{2}{n})^{n-1} \geq \frac{1}{2}e^{-2-2/n} > 0.06$ . So  $C_n < E_n$  at  $\varepsilon = 2/n$ , and Proposition 2.3 gives  $\varepsilon_n > 2/n$ . Hence  $x_n > 2 \geq 2(1-\varepsilon_n)$ . A computation shows that the hypothesis also holds for every  $8 \leq n < 1688$ . So Corollary 3.6 holds for every even  $n \geq 8$ , and below  $n = 1688$  it is computer-assisted. By Lemma 3.4 the law of  $A_n$  is a mixture of 2 unimodal laws with different modes, as is the limit law in [10, eq. (1)].

### 3.3 The bulk

Let  $\text{CP}_n$  be the law of  $\sum_{d=1}^{n-1} dN_d$ , where the  $N_d$  are independent Poisson variables with means  $x/(d(d+1))$ . On average  $x/(d(d+1))$  flipped vertices have a family of size  $d$ . If no 2 flips fall in the same family, which is likely when  $x^2 \log n/n$  is small, then  $M_n$  is the total size of the flipped families, and this is close to a compound Poisson sum.

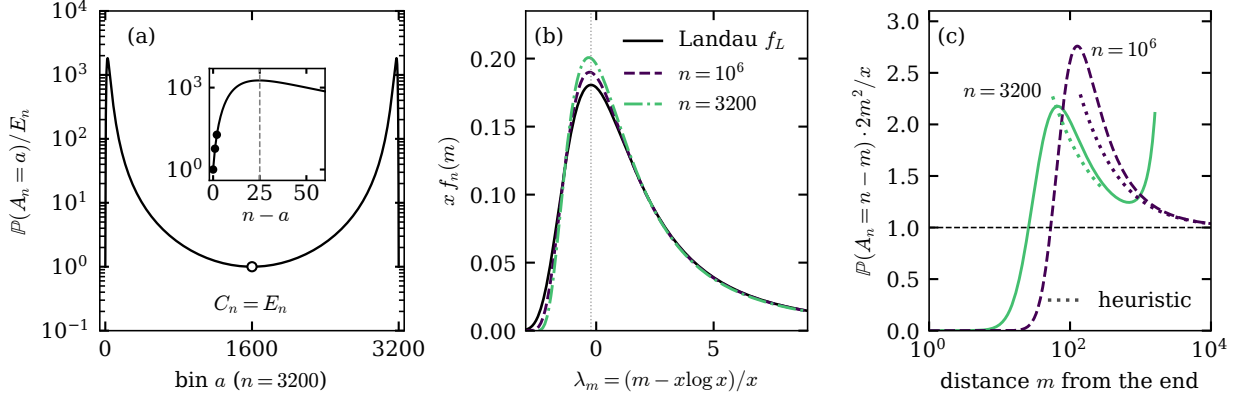


Figure 2: The elephant walk at its crossing. (a) The law of  $A_n$  at  $n = 3200$  and its crossing  $x = 11.5468$ , divided by the end  $E_n$ . The center equals the end, and the law rises from each end to a mode and falls to the center. The inset shows the last 60 bins, with dots at bins  $n$ ,  $n-1$  and  $n-2$ . The end is below its neighbor, and the modes are 25 steps from each end (dashed). (b) The law of  $M_n$ , scaled as  $x f_n(m)$ , against  $\lambda_m = (m - x \log x)/x$ , with the Landau density  $f_L$ . The dotted line is its mode  $\lambda_0$ . (c) The tail  $\mathbb{P}(A_n = n - m)$  divided by  $x/(2m^2)$ . The dashed line is the limit 1 of Theorem 3.8, and the dotted curves are the heuristic correction  $1 + x(2 \log m + 2\gamma - 3)/m$  for  $2x \log x \leq m \leq n/x$ . The rise of the  $n = 3200$  curve near  $m = n/2$  is the mirror term  $f_n(n - m)$ . At  $n = 10^6$  we use  $x = g_n = 22.4407$ .

*Remark 3.7* (compound Poisson and Landau limits, proofs omitted). For  $n \geq 2$  and  $0 < \varepsilon \leq \frac{1}{4}$ , Fourier inversion, the tree of Lemma 3.1 and the Efron–Stein inequality [20] give  $\sup_m |f_n(m) - \text{CP}_n(m)| \leq \delta_B := \frac{x^2 \log n}{n} + \frac{4x^2}{n} e^{4x^2/n} + \frac{4x^2(H_n^2 + H_n)}{n}$ . Let  $f_L$  be the Landau density, with  $\int e^{-u\lambda} f_L(\lambda) d\lambda = u^u$  for  $u > 0$ , and put  $\lambda_m = (m - x \log x)/x$ . Families of size  $d$  are flipped at rate about  $x/d^2$ , so the families of size up to about  $x$  add up to nearly  $x \log x$ , and the few families of size of order  $x$  give fluctuations of order  $x$ . Under the standing assumptions of this section, comparing characteristic functions then gives  $\sup_m |x f_n(m) - f_L(\lambda_m)| \leq C(\frac{1 + \log x}{x} + \frac{x}{n}) + x \delta_B$  with an absolute constant  $C$ . So  $(M_n - x \log x)/x$  tends to the Landau law when  $x \rightarrow \infty$  and  $x^3 \log^2 n/n \rightarrow 0$ , in particular at the crossing. We omit the details of these 2 bounds. At  $n = 10^6$  and  $x = 22.4407$  the bound  $\delta_B$  is still about 0.46, but there  $|f_n(m)/\text{CP}_n(m) - 1| \leq 3.5 \cdot 10^{-4}$  for  $1 \leq m \leq 10^4$ . At every tested crossing  $m^*$  equals the mode of  $\text{CP}_n$ . If  $f_L''(\lambda_0) < 0$  at the mode  $\lambda_0 = -0.2228$  of  $f_L$ , then the unimodality of  $f_L$  [19] and the bound above give  $m^* = x \log x + \lambda_0 x + O(\sqrt{x \log x})$  at the crossing for large  $n$ . We computed  $f_L''(\lambda_0) = -0.0789$  by quadrature but did not certify its sign.

### 3.4 The tail

**Theorem 3.8** (the tail). *Let  $n$  be even, and let  $x = n\varepsilon \geq 1$  depend on  $n$  with  $x \log^2 n/n \rightarrow 0$ . Let  $\omega_n \rightarrow \infty$  with  $\omega_n x \log^2(x+1) = o(n)$ , and let  $\delta'_n \rightarrow 0$ . Then*

$$f_n(m) = \frac{x}{m(m+1)}(1 + o(1)) \quad \text{and} \quad \mathbb{P}(A_n = n - m) = \frac{x}{2m^2}(1 + o(1)),$$

*uniformly for  $\omega_n x \log^2(x+1) \leq m \leq \delta'_n n$ .*

In the proof (Appendix B) the lower bound uses the first flipped family, which has size  $m$  with probability about  $x/(m(m+1))$  by Lemma B.4, while the other flips change the count

only by a relative amount of order  $x \log m/m$ . For the upper bound we show  $\mathbb{P}(M_n \geq m) \leq \frac{x}{m}(1 + O(x \log^2 m/m))$ , and the unimodality of  $f_n$  turns this into a local bound. The mirror term  $f_n(n-m)$  is at most  $C_n = O(x/n^2)$  by Lemma 3.4 and Theorem 2.6(b), so bin  $n-m$  is reached through a family of size  $m$  only when  $X_1 = +1$ , and this gives the constant  $\frac{1}{2}$ . At  $n = 10^6$ ,  $x = 22.4407$  and  $m = 10^4$  the exact ratio  $\mathbb{P}(A_n = n-m)/(x/m^2)$  is 0.51889. At moderate  $m$ ,  $\text{CP}_n$  suggests  $f_n(m) m(m+1)/x \approx 1 + x(2 \log m + 2\gamma - 3)/m$ , and this factor is still about 2 near  $m = 70$  when  $n = 1600$ .

### 3.5 The center

**Theorem 3.9** (shape at the crossing). *There are absolute constants  $C_w$  and  $n_0$  with the following property. Let  $n \geq n_0$  be even with  $x_n \in [2, n^{1/4}]$ . Then at the crossing  $\mathbb{P}(A_n = a) > C_n$  for every  $1 \leq a \leq n-1$  with  $|a-h| \geq C_w(n x_n \log n)^{1/2}$ .*

By Theorem 2.7,  $x_n \in [2, n^{1/4}]$  for large  $n$ , so at the crossing the window has width of order  $n^{1/2} \log n$ .

*Proof.* Write  $x = x_n$  and  $\varepsilon = \varepsilon_n$ , so that  $C_n = E_n$ . By symmetry we take  $a = n-m$  with  $1 \leq m \leq h$ , and put  $t = h-m$ . Since  $2 \leq x \leq n^{1/4}$ , every error term below is bounded by a function of  $n$  alone, and we take  $n$  large. Put  $\Pi(m) = \frac{1}{2} \left[ \frac{1}{m(m+1)} + \frac{1}{(n-m)(n-m+1)} \right]$ , so that  $x\Pi(h) = \frac{4\varepsilon}{n+2}$ . As  $\varepsilon(1+\log n) \rightarrow 0$ , Theorem 2.6(b) gives  $C_n \leq x\Pi(h)(1+\eta)$  with  $\eta = n^{2\varepsilon} e^{5200\varepsilon} - 1$ . Put  $u = \varepsilon(2 \log n + 5200)$ . Then  $u \leq \frac{1}{2}$  for large  $n$ , and  $e^u - 1 \leq 2u$  for  $0 \leq u \leq \frac{1}{2}$ , so  $\eta \leq 2u \leq 10^4 \varepsilon \log n$ .

If  $m \leq m_h$ , then  $f_n(m) \geq f_n(1)$  by Lemma 3.4. The proof of Theorem 3.5 gives  $f_n(1) \geq \frac{x}{2(1-\varepsilon)} f_n(0)$ , and  $f_n(0) = 2E_n = 2C_n$ . So  $\mathbb{P}(A_n = n-m) \geq \frac{1}{2} f_n(m) \geq \frac{x}{2(1-\varepsilon)} C_n > C_n$ .

If  $m_h < m \leq n/8$ , then  $f_n(m) \geq f_n(k)$  with  $k = \lfloor n/8 \rfloor$ , by Lemma 3.4. Proposition B.5(a) at  $k$  gives  $f_n(k) \geq \frac{x}{k(k+1)} (1 - O(\frac{x \log n}{n}))$ . So  $\mathbb{P}(A_n = n-m) \geq \frac{32x}{n(n+8)} (1 - o(1))$ , which exceeds  $\frac{4x}{n(n+2)} (1+\eta) \geq C_n$ .

If  $n/8 < m \leq h$ , then  $m$  and  $n-m$  are at least  $n/8$ , and Proposition B.5 at  $m$  and at  $n-m$  gives  $\mathbb{P}(A_n = n-m) \geq x\Pi(m)(1-\bar{e})$  with  $\bar{e} = O(x \log n/n)$ . Put  $N = n+1$  and  $s = 2t/N$ , so that  $m + \frac{1}{2} = \frac{N}{2}(1-s)$  and  $n-m + \frac{1}{2} = \frac{N}{2}(1+s)$ . Since  $\frac{1}{k(k+1)} \geq (k + \frac{1}{2})^{-2}$  and  $\Pi(h) = \frac{1}{h(h+1)} = (\frac{N}{2})^{-2} (1-N^{-2})^{-1}$ , we get  $\frac{\Pi(m)}{\Pi(h)} \geq \frac{1}{2} [(1-s)^{-2} + (1+s)^{-2}] (1-N^{-2}) \geq (1+3s^2)(1-N^{-2})$ , because  $\frac{1}{2} [(1-s)^{-2} + (1+s)^{-2}] = \sum_{k \geq 0} (2k+1)s^{2k}$ . With  $b = \bar{e} + N^{-2} \leq \frac{1}{2}$  this gives  $\mathbb{P}(A_n = n-m) \geq x\Pi(h)(1+3s^2)(1-b) \geq x\Pi(h)(1+\frac{3}{2}s^2-b)$ , which exceeds  $x\Pi(h)(1+\eta) \geq C_n$  as soon as  $\frac{3}{2}s^2 > \eta + b$ . Now  $\eta + b \leq A_2 x \log n/n$  for an absolute constant  $A_2$ , since  $x \geq 2$ , and  $s \geq t/n$ . So the condition holds when  $t \geq C_w(n x \log n)^{1/2}$  with  $C_w^2 > 2A_2/3$ .  $\square$

The window has this size because, at distance  $t$  from  $h$ , the main term  $x\Pi(m)$  exceeds its value at  $h$  by a factor of about  $1 + 12t^2/n^2$ , while we know  $C_n$  and the law only up to relative errors of order  $x \log n/n$ . We did not compute  $C_w$  or  $n_0$ . Inside the window the proof gives nothing, and closing it needs second differences of the law.

**Conjecture 3.10** (convexity between the humps). *For every  $n \geq 10$  and  $x = x_n$ , the second difference  $\mathbb{P}(A_n = a-1) - 2\mathbb{P}(A_n = a) + \mathbb{P}(A_n = a+1)$  is positive for  $2m^* + 1 \leq a \leq n - 2m^* - 1$ .*

With the symmetry of the law, the conjecture makes the law strictly convex on  $[2m^*, n - 2m^*]$ , so  $h$  is its unique minimizer there (the 2 central bins for odd  $n$ ). The next corollary shows that  $m^* = o(n)$  at the crossing.

**Corollary 3.11** (the modes). *For large even  $n$  the modes  $a^*$  of the law of  $A_n$  at the crossing satisfy  $\min(a^*, n - a^*) = O(x_n \log^2 x_n)$ .*

*Proof.* Write  $x = x_n$  and  $\varepsilon = \varepsilon_n$ . By Theorem 2.7,  $x \sim 2 \log n$ , so for large  $n$  the hypotheses of Theorem 2.6(b) and Lemma B.7 hold. The law is symmetric about  $h$ , so  $m^* = \min(a^*, n - a^*)$  maximizes  $f_n(m) + f_n(n - m)$  over  $0 \leq m \leq h$ . Lemma B.7 gives  $m_h \leq h$  for large  $n$ . So Lemma 3.4 and Theorem 2.6(b) give  $f_n(n - m) \leq f_n(h) = C_n = O(x/n^2)$  for  $m \leq h$ , and hence  $f_n(m^*) \geq f_n(m_h) - C_n$ . Let  $\ell$  be as in the proof of Lemma B.7. Then  $f_n(m_h) \geq f_n(\ell) \geq \frac{x}{2\ell(\ell+1)}$ , so  $f_n(m^*) \geq \frac{x}{4\ell(\ell+1)}$  for large  $n$ . If  $m^* > m' = \max(m_h, \ell)$ , then Lemmas B.1 and B.2 give  $(m^* - m' + 1)f_n(m^*) \leq \mathbb{P}(M_n \geq \ell) \leq xH_\ell/\ell$ . Hence  $m^* \leq \max(m_h, \ell) + 4(\ell + 1)H_\ell$  in every case. Finally Lemma B.7 gives  $m_h = O(x \log^2 x)$ , and  $\ell = O(x \log x)$  and  $H_\ell = O(\log x)$ .  $\square$

So for large even  $n$  Theorem 3.9 covers the bins outside  $[2m^*, n - 2m^*]$ , and the conjecture would make the center the unique interior minimum. The conjecture holds for every  $10 \leq n \leq 600$  and for  $n = 800, 1600, \dots, 25600$ . On the closed range  $[2m^*, n - 2m^*]$  it fails only at the 2 endpoints and only for some  $n \leq 51$ . Heuristically, a concave stretch of  $f_n$  far from its mode is added to the tail  $x/(2r^2)$  of the other branch, with  $r = n - a$ , and the second difference  $3x/r^4$  of this tail beats the concave curvature, which is of order  $x^2/(n^2r^3)$ . We leave open the window around the center, the tail between  $x \log x$  and  $x \log^2 x$ , and effective constants in Theorems 3.8 and 3.9.

## 4 Heavy-tailed runs

In this section the walk remembers the time since its last switch. The steps form maximal runs whose directions alternate. The first run has length  $D$ , and the later runs have i.i.d. lengths  $\xi_1, \xi_2, \dots$  distributed as  $\xi$ , where  $\bar{F}(k) = \mathbb{P}(\xi \geq k) = k^{-\alpha}$  for  $k \geq 1$  and some  $\alpha > 0$ . So  $p_k = \mathbb{P}(\xi = k) = k^{-\alpha} - (k+1)^{-\alpha} > 0$  for every  $k \geq 1$ . The first direction is fair, and  $D$ ,  $(\xi_i)$  and the first direction are independent. With a fresh start  $D$  has the law of  $\xi$ . For  $\alpha > 1$  the stationary start is  $\mathbb{P}(D = d) = \bar{F}(d)/\mu$ , where  $\mu = \mathbb{E}\xi = \zeta(\alpha)$ . This law is stationary, since the number of steps left in the current run (counting the current one) is a Markov chain that moves from  $k \geq 2$  to  $k - 1$  and from 1 to a fresh copy of  $\xi$ , and  $\bar{F}(k) = \bar{F}(k+1) + p_k \bar{F}(1)$  makes  $\bar{F}(k)/\mu$  invariant. We define  $A_n$ ,  $C_n$  and  $E_n$  as in Section 1, put  $R_n = E_n/C_n$ , and write  $\mathbb{P}_+$  for the law given that the first run is  $+$ .

**Proposition 4.1** (the edge). *For every law of  $D$ ,  $E_n = \frac{1}{2}\mathbb{P}(D \geq n)$ . With a fresh start  $E_n = \frac{1}{2}n^{-\alpha}$ . With the stationary start  $E_n = \zeta(\alpha, n)/(2\zeta(\alpha)) = n^{1-\alpha}(1 + \theta_n(\alpha - 1)/n)/(2(\alpha - 1)\zeta(\alpha))$  for some  $\theta_n \in [0, 1]$ , where  $\zeta(\alpha, n) = \sum_{k \geq n} k^{-\alpha}$  is the Hurwitz zeta function.*

*Proof.* All  $n$  steps are  $+1$  exactly when the first direction is  $+$  and  $D \geq n$ . Hence  $E_n = \frac{1}{2}\mathbb{P}(D \geq n)$ , and  $\mathbb{P}(\xi \geq n) = n^{-\alpha}$ . For the stationary start  $\mathbb{P}(D \geq n) = \zeta(\alpha, n)/\zeta(\alpha)$ . Since  $x^{-\alpha}$  is decreasing,  $\int_n^\infty x^{-\alpha} dx \leq \zeta(\alpha, n) \leq n^{-\alpha} + \int_n^\infty x^{-\alpha} dx$ , and the integral equals  $n^{1-\alpha}/(\alpha - 1)$ .  $\square$

So the edge only sees the first run, while the center depends on all the runs. Let  $\tau_m = \xi_1 + \dots + \xi_m$  with  $\tau_0 = 0$ ,  $u_m(k) = \mathbb{P}(\tau_m = k)$  and  $q_m(k) = \mathbb{P}(\tau_{m-1} < k \leq \tau_m) = \sum_{j \geq 1} \bar{F}(j)u_{m-1}(k - j)$  for  $m, k \geq 1$ . For the delayed versions let  $D$  be independent of  $(\xi_i)$ , and put  $u_m^D(k) = \mathbb{P}(D + \tau_{m-1} = k)$ ,  $q_1^D(k) = \mathbb{P}(D \geq k)$  and  $q_m^D(k) = \mathbb{P}(D + \tau_{m-2} < k \leq D + \tau_{m-1})$  for  $m \geq 2$ . For the stationary start  $u_m^D(k) = \sum_d \frac{\bar{F}(d)}{\mu} u_{m-1}(k - d) = q_m(k)/\mu$ .

The  $+$  runs and the  $-$  runs form 2 independent renewal processes. The walk is in bin  $a$  at time  $n$  exactly when one of them sits at its level and the other is inside a run that covers its level.

**Proposition 4.2** (two renewal processes). *Let  $D$  have any law on  $\{1, 2, \dots\}$ . For  $a + b = n$  with  $a, b \geq 1$ ,*

$$\mathbb{P}_+(A_n = a) = \sum_{m \geq 1} \left[ u_m(b) q_{m+1}^D(a) + u_m^D(a) q_m(b) \right],$$

*and  $\mathbb{P}_+(A_n = n) = \mathbb{P}(D \geq n)$ . In particular, for even  $n$ ,  $C_n = \sum_{m \geq 1} [u_m(h) q_{m+1}^D(h) + u_m^D(h) q_m(h)]$ . For the fresh start this is  $C_n = \sum_m u_m(h) [q_m(h) + q_{m+1}(h)]$ .*

*Proof.* Let the  $+$  runs have lengths  $D, \xi_2^+, \xi_3^+, \dots$  and the  $-$  runs  $\xi_1^-, \xi_2^-, \dots$ , and let  $s_j^\pm$  be the sum of the first  $j$  lengths of each sign, with  $s_0^\pm = 0$ . So  $s_j^+$  has law  $u_j^D$  and  $s_j^-$  has law  $u_j$ , and the 2 sequences are independent. The  $j$ -th  $+$  run covers the steps in  $(s_{j-1}^+ + s_{j-1}^-, s_j^+ + s_{j-1}^-]$ , and the  $j$ -th  $-$  run covers  $(s_j^+ + s_{j-1}^-, s_j^+ + s_j^-]$ . If step  $n = a + b$  is in the first run then  $A_n = n$ , which happens exactly when  $D \geq n$ . If it is in the  $j$ -th  $+$  run with  $j \geq 2$ , then  $A_n = n - s_{j-1}^-$ , so  $A_n = a$  means  $s_{j-1}^- = b$  and  $s_{j-1}^+ < a \leq s_j^+$ , which has probability  $u_{j-1}(b) q_j^D(a)$ . If it is in the  $j$ -th  $-$  run, then  $A_n = s_j^+$ , so  $A_n = a$  means  $s_j^+ = a$  and  $s_{j-1}^- < b \leq s_j^-$ , which has probability  $u_j^D(a) q_j(b)$ . Summing over these disjoint cases, with  $m = j - 1$  in the first and  $m = j$  in the second, gives the formula. For even  $n$  the symmetry gives  $C_n = \frac{1}{2} [\mathbb{P}_+(A_n = h) + \mathbb{P}_+(A_n = n - h)] = \mathbb{P}_+(A_n = h)$ .  $\square$

This identity is the coefficient form of the double transform of Godrèche and Luck [11, Section 7] for the occupation time of an alternating renewal process.

**Lemma 4.3** (stable and local limits). *Let  $1 < \alpha < 2$  and  $a_m = m^{1/\alpha}$ . Put  $K_\alpha = \Gamma(2 - \alpha)/(\alpha - 1) = -\Gamma(1 - \alpha) > 0$  and  $\sigma_\alpha = 2K_\alpha |\cos \frac{\pi\alpha}{2}|$ .*

- (i)  $\mathbb{E}e^{i\theta\xi} = 1 + i\mu\theta + K_\alpha(-i\theta)^\alpha + O(\theta^2)$  as  $\theta \rightarrow 0$ , with the principal branch.
- (ii)  $(\tau_m - m\mu)/a_m$  converges in law to a variable  $G$  with  $\mathbb{E}e^{-sG} = e^{K_\alpha s^\alpha}$  for  $s \geq 0$  and  $\mathbb{E}e^{i\theta G} = \exp(K_\alpha(-i\theta)^\alpha)$ . The law of  $G$  is totally skewed to the right and has a bounded, uniformly continuous density  $g$  with  $g(z) \rightarrow 0$  as  $|z| \rightarrow \infty$ .
- (iii) (Gnedenko)  $\sup_k |a_m u_m(k) - g((k - m\mu)/a_m)| \rightarrow 0$ , and  $\bar{u} := \sup_{m \geq 1} \sup_k a_m u_m(k) < \infty$ .
- (iv) If  $G'$  is an independent copy of  $G$ , then  $G - G'$  is symmetric  $\alpha$ -stable with  $\mathbb{E}e^{i\theta(G - G')} = e^{-\sigma_\alpha |\theta|^\alpha}$ , and  $I_\alpha := \int_{\mathbb{R}} g(z)^2 dz = \Gamma(1 + 1/\alpha)/(\pi\sigma_\alpha^{1/\alpha})$ .

*Proof.* (i) Summation by parts gives  $1 - \mathbb{E}x^\xi = (1 - x) \sum_{k \geq 1} \bar{F}(k) x^{k-1}$  for  $|x| \leq 1$ . With  $x = e^{-u}$  and  $\text{Li}_\alpha(w) = \sum_{k \geq 1} k^{-\alpha} w^k$  this is  $1 - \mathbb{E}e^{-u\xi} = (e^u - 1) \text{Li}_\alpha(e^{-u})$ . For  $\text{Re } u \geq 0$  and  $0 < |u| < 2\pi$ , [21, 25.12.12] gives  $\text{Li}_\alpha(e^{-u}) = \Gamma(1 - \alpha) u^{\alpha-1} + \zeta(\alpha) + O(|u|)$ . So  $1 - \mathbb{E}e^{-u\xi} = \mu u + \Gamma(1 - \alpha) u^\alpha + O(|u|^2)$ , and  $u = -i\theta$  gives (i). (ii) By (i),  $e^{-i\mu t} \mathbb{E}e^{it\xi} = 1 + K_\alpha(-i\theta)^\alpha/m + O(\theta^2 m^{-2/\alpha})$  for  $t = \theta/a_m$ . As  $2/\alpha > 1$ , the characteristic function of  $(\tau_m - m\mu)/a_m$  tends to  $\exp(K_\alpha(-i\theta)^\alpha)$ , whose modulus  $\exp(-\frac{1}{2}\sigma_\alpha |\theta|^\alpha)$  is integrable. Lévy's continuity theorem, Fourier inversion and the Riemann–Lebesgue lemma give  $G$  and  $g$  (the domain of attraction also follows from [17, Theorem XVII.5.2]). For real  $s > 0$ , the expansion of  $1 - \mathbb{E}e^{-u\xi}$  above with  $u = s/a_m$  gives  $\mathbb{E}e^{-s(\tau_m - m\mu)/a_m} \rightarrow e^{K_\alpha s^\alpha}$  in the same way, and with  $2s$  in place of  $s$  this gives uniform integrability, so  $\mathbb{E}e^{-sG} = e^{K_\alpha s^\alpha}$ . (iii) Every  $p_k$  is positive, so  $\xi$  has maximal span 1, and Gnedenko's local limit theorem [16, Chapter 4] gives the limit. Then  $a_m u_m(k) \leq \|g\|_\infty + 1$  for large  $m$  and  $a_m u_m(k) \leq a_m$  for the others, so  $\bar{u} < \infty$ . (iv) The function  $|\mathbb{E}e^{i\theta G}|^2 = e^{-\sigma_\alpha |\theta|^\alpha}$  is the characteristic function of  $G - G'$ , and Plancherel's theorem gives  $I_\alpha = \frac{1}{2\pi} \int e^{-\sigma_\alpha |\theta|^\alpha} d\theta$ , which is  $\Gamma(1 + 1/\alpha)/(\pi\sigma_\alpha^{1/\alpha})$ .  $\square$

**Theorem 4.4** (the center for  $1 < \alpha < 2$ ). *Let  $1 < \alpha < 2$  and let  $D$  be any a.s. finite first run, in particular either start. As  $n \rightarrow \infty$  over even  $n$ ,*

$$C_n \sim 2I_\alpha\left(\frac{\mu}{h}\right)^{1/\alpha} = \frac{2\Gamma(1+1/\alpha)}{\pi} \left(\frac{\mu}{K_\alpha|\cos(\pi\alpha/2)|n}\right)^{1/\alpha}.$$

Both kinds of runs must fill about  $h$  steps, which takes about  $h/\mu$  runs of each kind. Each renewal process is then within about  $n^{1/\alpha}$  of its mean, so the 2 meet at a given level with probability of order  $n^{-1/\alpha}$ . In the proof (Appendix C) we split the sum in Proposition 4.2 by the number  $m$  of runs. With  $b_h = (h/\mu)^{1/\alpha}$ , the terms with  $|m\mu - h| \leq Mb_h$  form a Riemann sum for  $2b_h^{-1} \int_{-M}^M g^2$  by Lemma 4.3(iii). The terms with  $m \leq h/(4\mu)$  add up to  $O(h^{-\alpha}) = o(b_h^{-1})$ , since a large sum of few runs is most likely made by one long run. The others add up to  $O(b_h^{-1})\mathbb{P}(|G| \geq M-1) + o(b_h^{-1})$ , and we let  $M \rightarrow \infty$ .

With a stationary start the first run covers all  $n$  steps with probability of order  $n^{1-\alpha}$ , and this is comparable with  $C_n$  when  $1 - \alpha = -1/\alpha$ , that is  $\alpha^2 - \alpha - 1 = 0$ .

**Corollary 4.5** (the golden-ratio exponent). *Let  $1 < \alpha < 2$  and  $e(\alpha) = 1 - \alpha + 1/\alpha$ . If  $\mathbb{P}(D \geq n) \sim c_D n^{-\beta_D}$  with  $c_D, \beta_D > 0$ , then  $R_n \sim c_D n^{1/\alpha - \beta_D} / (4I_\alpha(2\mu)^{1/\alpha})$ . In particular  $R_n \sim C(\alpha)n^{e(\alpha)}$  for the stationary start and  $R_n \sim C_{\text{fr}}(\alpha)n^{e(\alpha)-1}$  for the fresh start, where*

$$C(\alpha) = \frac{\pi(K_\alpha|\cos \frac{\pi\alpha}{2}|)^{1/\alpha}}{4(\alpha-1)\Gamma(1+\frac{1}{\alpha})\zeta(\alpha)^{1+1/\alpha}}, \quad C_{\text{fr}}(\alpha) = (\alpha-1)\zeta(\alpha)C(\alpha).$$

Since  $e(\alpha) > 0$  exactly when  $\alpha < \varphi$ , with a stationary start the ends win for all large  $n$  if  $\alpha < \varphi$ , and the center wins if  $\alpha > \varphi$ . At  $\alpha = \varphi$ ,

$$R_n \rightarrow C(\varphi) = \frac{\pi\varphi(\varphi\Gamma(\varphi^{-2})|\cos \frac{\pi\varphi}{2}|)^{1/\varphi}}{4\Gamma(\varphi)\zeta(\varphi)^\varphi} = 0.7761317206\dots,$$

so the center still wins by the factor  $1/C(\varphi) = 1.2884$ . With a fresh start  $e(\alpha) - 1 < 0$ , and the center wins for every  $1 < \alpha < 2$ .

*Proof.* By Proposition 4.1 and Theorem 4.4, with  $h = n/2$ , we have  $R_n = \mathbb{P}(D \geq n)/(2C_n)$ , and this is asymptotic to  $c_D n^{-\beta_D} / (4I_\alpha(2\mu/n)^{1/\alpha})$ , which is the general formula. By Lemma 4.3(iv),  $4I_\alpha(2\mu)^{1/\alpha} = 4\Gamma(1+\frac{1}{\alpha})\mu^{1/\alpha} / (\pi(K_\alpha|\cos \frac{\pi\alpha}{2}|)^{1/\alpha})$ . The fresh start has  $c_D = 1$  and  $\beta_D = \alpha$ , which gives  $C_{\text{fr}}(\alpha)$ . By Proposition 4.1 the stationary start has  $c_D = 1/((\alpha-1)\mu)$  and  $\beta_D = \alpha-1$ , which gives  $C(\alpha)$ . Next,  $e(\alpha) = -(\alpha^2 - \alpha - 1)/\alpha$ , and  $\varphi$  is the positive root of  $\alpha^2 - \alpha - 1$ . At  $\alpha = \varphi$  we have  $\alpha - 1 = 1/\varphi$ ,  $1 + 1/\alpha = \varphi$  and  $2 - \alpha = \varphi^{-2}$ . So  $K_\varphi = \varphi\Gamma(\varphi^{-2})$ , and substituting these into  $C(\alpha)$  gives the closed form.  $\square$

Corollary 4.5 is the lattice form, with its constant, of the heuristic of Wang, Li and Hänggi [15, Section 2.1]. The golden ratio also appears for the elephant walk, as the log-concavity threshold  $1/\varphi$  of [10, Proposition 2.2], but we know of no connection between the 2. The approach to  $C(\varphi)$  is slow. At  $\alpha = \varphi$  we have  $R_{10^5}/C(\varphi) = 0.938$ , and  $R_n$  increases on the computed grid (Table 2 and Figure 3).

*Remark 4.6* (other exponents). Table 3 collects the cases that Corollary 4.5 leaves out. For  $\alpha > 2$  let  $v = \text{Var } \xi = 2\zeta(\alpha-1) - \zeta(\alpha) - \zeta(\alpha)^2$  and  $C_2(\alpha) = \sqrt{\pi v/(2\mu)}/(2(\alpha-1)\mu)$ . The proof of Theorem 4.4 works with the Gaussian local limit theorem and  $a_m = \sqrt{vm}$ , and for  $\alpha = 2$  with the norming  $a_m \sim \sqrt{m \log m}$  of [17, Theorem XVII.5.3 and eq. (XVII.5.23)]. In these 2 rows the

| $\alpha$  | $e(\alpha)$ | $C(\alpha)$ | $C_{\text{fr}}(\alpha)$ | $R_{1600}$         | $R_{10^4}$         | $R_{10^5}$         |
|-----------|-------------|-------------|-------------------------|--------------------|--------------------|--------------------|
| 1.3       | +0.469231   | 0.422717    | 0.498631                | 17.7593 [1.3180]   | 38.7537 [1.2171]   | 106.149 [1.1317]   |
| 1.5       | +0.166667   | 0.647971    | 0.846371                | 2.14409 [0.9675]   | 2.9528 [0.9818]    | 4.37642 [0.9914]   |
| $\varphi$ | 0           | 0.776132    | 1.073675                | 0.661763 [0.8526]  | 0.697331 [0.8985]  | 0.727689 [0.9376]  |
| 5/3       | -1/15       | 0.835101    | 1.182237                | 0.411815 [0.8064]  | 0.389307 [0.8614]  | 0.352822 [0.9102]  |
| 1.7       | -0.111765   | 0.880046    | 1.265508                | 0.298369 [0.7733]  | 0.262005 [0.8334]  | 0.215839 [0.8881]  |
| 1.9       | -0.373684   | 1.438986    | 2.266075                | 0.0449706 [0.4923] | 0.0256505 [0.5569] | 0.0121612 [0.6242] |

Table 2: Constants of Corollary 4.5 and exact values of  $R_n = E_n/C_n$  for the stationary start, with  $R_n/(C(\alpha)n^{e(\alpha)})$  in brackets.

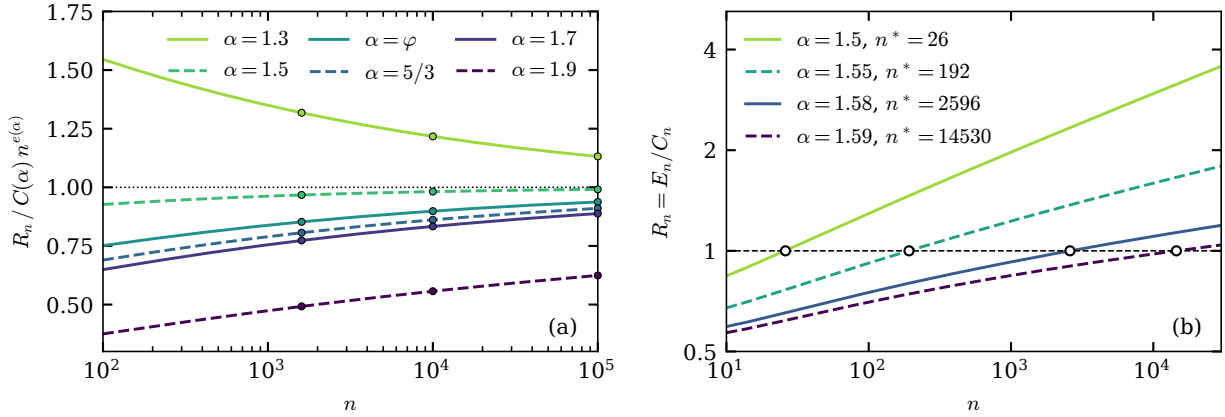


Figure 3: Alternating runs with a stationary start. (a)  $R_n/(C(\alpha)n^{e(\alpha)})$  for even  $n$  from 100 to  $10^5$ , with the entries of Table 2 as dots. (b)  $R_n$  for  $\alpha = 1.5, 1.55, 1.58, 1.59$  below  $\varphi$ . Each curve crosses 1 once on  $10 \leq n \leq 3 \cdot 10^4$ , at  $n^*(\alpha)$  (circles).

center formula holds for every a.s. finite first run, so a fresh start also gives  $R_n \rightarrow 0$ . For  $\alpha = 1$  the same proof with a skewed Cauchy limit and the centering  $m(H_{m+1} - 1)$  suggests the row of the table, where  $\tilde{m}_h$  solves  $m(H_{m+1} - 1) = h$ . For  $0 < \alpha < 1$  only the fresh start makes sense, and  $\frac{2}{\pi} \tan \frac{\pi\alpha}{2}$  is the density at  $\frac{1}{2}$  of Lamperti's occupation-time law [12, 11]. Here the proof splits the sum in Proposition 4.2 on the scale of  $\log m$  and uses the moments of a positive stable law [18, Theorem 2.6.3] and the series for its density [17, Lemma XVII.6.1]. We omit the details for  $\alpha \geq 2$  and  $0 < \alpha < 1$ . So with a fresh start the threshold is  $\alpha = 1$ .

*Remark 4.7* (crossover sizes and heat conduction). For  $\alpha < \varphi$  and a stationary start the crossover is late when  $\alpha$  is close to  $\varphi$ . For  $\alpha = 1.5, 1.55, 1.58, 1.59$  and every even  $n$  in  $[10, 3 \cdot 10^4]$ ,  $R_n - 1$  changes sign once, at  $n^*(\alpha) = 26, 192, 2596, 14530$ . Heuristically,  $C(\alpha)n^{e(\alpha)} = 1$  gives  $\log n^*(\alpha) = 0.1834/(\varphi - \alpha) + O(1)$  as  $\alpha \uparrow \varphi$ . At the value  $\alpha = 5/3$  of Lévy-walk models of heat conduction [14, 13],  $R_n \sim C(5/3)n^{-1/15}$  with  $C(5/3) = 0.8351$ .

*Remark 4.8* (aging). Let the walk take a fair first step and then switch direction at step  $k \geq 2$  with probability  $c/k$ , where  $0 < c < 2$ . Then  $A_n/n$  converges in law to Beta( $c, c$ ) [22, Example 1.1 and Theorem 1.4] (their chain is this walk for  $c \leq 1$ ), [23, Theorem 1], and the limit is uniform exactly when  $c = 1$  [23, Remark 2]. At finite  $n$  the first step matters. Let  $\mathbb{P}_+$  be the law given that the first step is  $+1$ , and let  $X_n$  be the last step. At  $c = 1$  induction on  $n \geq 2$  gives  $\mathbb{P}_+(A_n = s, X_n = +1) = \frac{s-1}{n(n-1)}$  and  $\mathbb{P}_+(A_n = s, X_n = -1) = \frac{n-s}{n(n-1)}$  for  $1 \leq s \leq n$ , so  $A_n$  is uniform on  $\{1, \dots, n\}$  (after a shift of one step this is the exercise in [24, footnote 1 on

| $\alpha$         | start      | center $C_n$                                                    | ratio $R_n$                                                            | status                 |
|------------------|------------|-----------------------------------------------------------------|------------------------------------------------------------------------|------------------------|
| $\alpha > 2$     | stationary | $\sim \sqrt{2\mu/(\pi v n)}$                                    | $\sim C_2(\alpha)n^{3/2-\alpha}$                                       | same argument, omitted |
| $\alpha = 2$     | stationary | $\sim \sqrt{2\mu/(\pi n \log n)}$                               | $\sim \sqrt{\pi \log n / (8\zeta(2)^3 n)}$                             | same argument, omitted |
| $1 < \alpha < 2$ | fresh      | $\sim 2I_\alpha(\mu/h)^{1/\alpha}$                              | $\sim C_{\text{fr}}(\alpha)n^{e(\alpha)-1} \rightarrow 0$              | Corollary 4.5          |
| $\alpha = 1$     | fresh      | $\approx 2/(\pi^2 \tilde{m}_h)$                                 | $\approx \pi^2 \tilde{m}_h / (8h) \sim \pi^2 / (8 \log n)$             | heuristic              |
| $0 < \alpha < 1$ | fresh      | $\sim \frac{2}{\pi} \tan \frac{\pi\alpha}{2} \cdot \frac{1}{n}$ | $\sim \frac{\pi}{4 \tan(\pi\alpha/2)} n^{1-\alpha} \rightarrow \infty$ | proof omitted          |

Table 3: Outcomes of Remark 4.6. The center wins when  $R_n \rightarrow 0$  and the ends win when  $R_n \rightarrow \infty$ .

p. 6]). Compared with no switch, a switch at time  $t$  multiplies the probability of a path by  $c/(t-c)$ , which increases in  $c$ , and every path with  $A_n < n$  switches. So by the argument of Proposition 2.3,  $\mathbb{P}_+(A_n = s)/\mathbb{P}_+(A_n = n)$  increases strictly in  $c$  for  $1 \leq s \leq n-1$ , and the fixed-start crossing is exactly  $c = 1$  for every  $n$  and every bin. With a fair first step the ends are  $\frac{1}{2n}$  at  $c = 1$  and every interior bin is  $\frac{1}{n}$ . By the same argument  $C_n/E_n$  increases strictly in  $c$ , from 0 as  $c \downarrow 0$  to 2 at  $c = 1$ , so the crossing  $c_n$  is unique and below 1. For even  $n$ ,  $c_n = 1 - \log 2 / (\log n + \gamma + 2 - 2 \log 2) + O((\log n)^{-3})$ , and the exact crossing at  $n = 200$  is  $c_{200} = 0.8925$ , against 0.8932 from the first 2 terms. This follows from  $E_n = \frac{1}{2} \prod_{k=2}^n (1 - c/k)$  and a local limit theorem with a rate,  $nC_n \rightarrow 4^{1-c}/B(c, c)$  uniformly for  $c$  near 1, and we omit the details. The exact statements here are also proved in Lean (Section 5).

We do not know a rate in Theorem 4.4, whether  $R_n$  is monotone with a single sign change of  $R_n - 1$ , how to make the case  $\alpha = 1$  rigorous, or the full shape of the law of  $A_n$ .

## 5 Verification

All computations are in Python, and the main ones were done by 2 independent methods. For example, a forward recursion for the law of  $M_n$  and a backward recursion for  $V_s$  agree to about  $10^{-14}$  at  $n = 12800$ . The proofs use computation in only 3 places. Some constants are values of explicit sums and integrals, Remark 3.7 uses a computed sign, and Corollary 3.6 holds for every even  $n \geq 8$  but for  $n < 1688$  its hypothesis rests on a computation of  $C_n/E_n$  at  $\varepsilon = 2/(n+2)$ . Conjecture 3.10, the tables, the figures and the numbers quoted outside the proofs rest on computation alone. Before each new computation we wrote down a prediction and a condition that would refute it. Of these predictions 8 failed, and none of them concerns a statement that we call a theorem. The scripts, their output and the prediction ledgers are in the Problem 131 folder of <https://github.com/apemmm/Kagey-Problems>, where SUPPLEMENT.md says what each script checks and `python -B verify_all.py --paper d` reruns them in a few minutes. The exact and finite statements, among them Proposition 2.3, Lemma 3.4, Theorem 3.5 and the identities behind Corollary 4.5, are also proved in Lean 4 in lean/PaperD, and MAP.md there lists each one with its Lean name.

**Acknowledgment.** The Lean code for this paper was generated by AI. AI was also used to revise some of the writing and to help cross-verify the code.

## A Proof of Theorem 2.6

Throughout,  $n \geq 2$  is even and we condition on  $X_1 = +1$ . Let  $\mathcal{L}_s F(b) = (1 - r_s(b))F(b) + r_s(b)F(b+1)$ , so that  $V_s = \mathcal{L}_s V_{s+1}$ . Both  $M_t$  and  $t - M_t \geq 1$  are nondecreasing. So from  $b \geq 1$

the chain stays at states  $\geq 1$ , and for  $b \geq 1$  we have  $V_s(b) = 0$  unless  $b$  is feasible, that is,  $b \in \mathcal{S}_s = \{b \in \mathbb{Z} : 1 \leq b \leq h, 1 \leq s - b \leq h\}$ . Note that  $\mathcal{S}_n = \{h\}$ . Theorem 2.6(a) follows from Lemma A.1(d), and (b) from Proposition A.4.

For  $c \geq 0$  the Pólya( $c$ ) chain moves at time  $s$  from  $b$  to  $b + 1$  with probability  $p_s^c(b) = \frac{b+c}{s+2c}$ . It is Pólya's urn with weights  $b + c$  and  $s - b + c$  for the 2 colors. For  $b + c > 0$  and  $s - b + c > 0$  let  $\Phi_s^c(b) = \binom{n-s}{h-b} B(h+c, h+c) / B(b+c, s-b+c)$ , where  $B$  is the beta function and the binomial coefficient is 0 unless  $0 \leq h - b \leq n - s$ .

**Lemma A.1.** *Let  $c \geq 0$ ,  $1 \leq s \leq n$ ,  $b + c > 0$  and  $s - b + c > 0$ .*

- (a)  $\Phi_s^c(b)$  is the probability that the Pólya( $c$ ) chain at  $b$  at time  $s$  is at  $h$  at time  $n$ . So  $\Phi_n^c(b) = \mathbf{1}\{b = h\}$ , and  $\Phi_s^c(b) = p_s^c(b)\Phi_{s+1}^c(b+1) + (1 - p_s^c(b))\Phi_{s+1}^c(b)$  for  $s < n$ .
- (b) If  $s < n$  and  $\Phi_s^c(b) > 0$ , then  $p_s^c(b)\Phi_{s+1}^c(b+1)/\Phi_s^c(b) = \bar{r}_s(b) := \frac{h-b}{n-s}$ .
- (c) If  $0 < \varepsilon < \frac{1}{2}$  and  $\beta = \frac{\varepsilon}{1-2\varepsilon}$ , then  $r_s(b) = p_s^{\beta s}(b)$  for  $0 \leq b \leq s$ .
- (d) If  $0 < \varepsilon < \frac{1}{2}$  and  $0 \leq c \leq \beta s$ , then  $V_s(b) \geq \Phi_s^c(b)$  for  $1 \leq b \leq s - 1$ . In particular  $V_{t+1}(1) \geq \Phi_{t+1}^0(1) = \pi_t$  for  $1 \leq t \leq h$ .

*Proof.* (a) Let  $x = b + c$ ,  $y = s - b + c$  and  $(x)_k = x(x+1) \cdots (x+k-1)$ . As in the proof of Lemma 2.4, each order of the  $n - s$  draws with  $h - b$  draws of the first color has probability  $(x)_{h-b}(y)_{n-s-h+b}/(x+y)_{n-s} = B(h+c, h+c)/B(x, y)$ , and there are  $\binom{n-s}{h-b}$  such orders. The recursion is the Markov property.

(b) Use  $\binom{n-s-1}{h-b-1}/\binom{n-s}{h-b} = \frac{h-b}{n-s}$  and  $B(x, y)/B(x+1, y) = \frac{x+y}{x} = 1/p_s^c(b)$ .

(c) Since  $1 + 2\beta = 1/(1 - 2\varepsilon)$ , we have  $\frac{b+\beta s}{s+2\beta s} = (1 - 2\varepsilon)(\frac{b}{s} + \beta) = r_s(b)$ .

(d) Let  $F_u = \Phi_u^c$  on  $\{1, \dots, u-1\}$  for  $s \leq u \leq n$ . Then  $F_u$  vanishes off  $\mathcal{S}_u$  and  $F_n(b) = \mathbf{1}\{b = h\}$ . By (a) and (c),  $\mathcal{L}_u F_{u+1}(b) - F_u(b)$  is the product of  $\frac{u}{2} - b$ ,  $\frac{1}{u+2c} - \frac{1}{u(1+2\beta)} \geq 0$  and  $F_{u+1}(b+1) - F_{u+1}(b)$ . Let  $b \in \mathcal{S}_u$ . If  $b = h$  then  $b \geq u/2$  and  $F_{u+1}(b+1) = 0$ . If  $u - b = h$  then  $b \leq u/2$  and  $F_{u+1}(b) = 0$ . Otherwise  $F_{u+1}(b+1)/F_{u+1}(b) = \frac{(h-b)(u-b+c)}{(h-u+b)(b+c)}$ , which is  $\geq 1$  exactly when  $b \leq u/2$ . So  $\mathcal{L}_u F_{u+1} \geq F_u$  on  $\mathcal{S}_u$ , and Lemma A.2(b) gives  $V_s \geq F_s$ . Finally  $\Phi_{t+1}^0(1) = \beta_{t+1}(h) = \pi_t$  by (a) and Lemma 2.4.  $\square$

**Lemma A.2.** *Let  $1 \leq s_0 \leq n$ , and for  $s_0 \leq s \leq n$  let  $F_s \geq 0$  be a function on  $\{1, \dots, s-1\}$ .*

- (a) If  $F_n(h) \geq 1$  and  $\mathcal{L}_s F_{s+1} \leq K_s F_s$  on  $\mathcal{S}_s$  for  $s_0 \leq s < n$ , with constants  $K_s \geq 0$ , then  $V_s \leq (\prod_{u=s}^{n-1} K_u) F_s$  on  $\mathcal{S}_s$ .
- (b) If  $F_n(b) \leq \mathbf{1}\{b = h\}$ , each  $F_s$  vanishes off  $\mathcal{S}_s$ , and  $\mathcal{L}_s F_{s+1} \geq F_s$  on  $\mathcal{S}_s$  for  $s_0 \leq s < n$ , then  $V_s \geq F_s$ .

*Proof.* We use backward induction on  $s$ , from  $V_n(b) = \mathbf{1}\{b = h\}$ . In (a),  $V_{s+1}$  vanishes at infeasible states and  $F_{s+1} \geq 0$ , so on  $\mathcal{S}_s$  we get  $V_s = \mathcal{L}_s V_{s+1} \leq \prod_{u>s} K_u \mathcal{L}_s F_{s+1} \leq \prod_{u \geq s} K_u F_s$ . In (b) we get  $V_s \geq \mathcal{L}_s F_{s+1} \geq F_s$  on  $\mathcal{S}_s$ , and  $F_s = 0 \leq V_s$  off  $\mathcal{S}_s$ .  $\square$

Next we build the supersolution. Let  $\psi$  be the digamma function and  $H_k = \sum_{j \leq k} 1/j$ .

**Lemma A.3.** *Let  $2 \leq s \leq n$ ,  $b \in \mathcal{S}_s$  and  $\tau \geq 0$ . Put  $X = b + \tau$ ,  $Y = s - b + \tau$ ,  $\Sigma = X + Y$  and  $z = (s - 2b)^2 / \Sigma^2$ . Then*

$$\partial_\tau \log \Phi_s^\tau(b) < -\log(1 - z) + \frac{1 + z}{(1 - z)\Sigma} + \frac{2(1 + z)}{3\Sigma^2(1 - z)^2}.$$

*Proof.* Since  $\log B(x, y) = \log \Gamma(x) + \log \Gamma(y) - \log \Gamma(x + y)$ , the derivative is  $2\psi(\Sigma) - \psi(X) - \psi(Y) - 2\psi(n + 2\tau) + 2\psi(h + \tau)$ . For  $x > 0$ , [21, 5.9.15] writes  $\log x - \frac{1}{2x} - \psi(x)$  as  $2 \int_0^\infty \frac{t dt}{(t^2 + x^2)(e^{2\pi t} - 1)}$ , which lies in  $(0, \frac{1}{12x^2})$  since  $\int_0^\infty \frac{t dt}{e^{2\pi t} - 1} = \frac{1}{24}$ . We use this upper bound for  $\psi(\Sigma)$  and lower bound for  $\psi(X)$  and  $\psi(Y)$ . The duplication formula [21, 5.5.5] gives  $2\psi(2y) = \psi(y) + \psi(y + \frac{1}{2}) + 2\log 2$ , and  $\psi$  is increasing, so  $2\psi(n + 2\tau) - 2\psi(h + \tau) > 2\log 2$ . Hence the derivative is less than  $\log \frac{\Sigma^2}{4XY} + (\frac{1}{2X} + \frac{1}{2Y} - \frac{1}{\Sigma}) + \frac{1}{12}(\frac{1}{X^2} + \frac{1}{Y^2})$ . Now  $4XY = \Sigma^2(1 - z)$  and  $X^2 + Y^2 = \Sigma^2(1 + z)/2$ , so the 3 terms are  $-\log(1 - z)$ ,  $\frac{X^2 + Y^2}{2XY\Sigma} = \frac{1+z}{(1-z)\Sigma}$  and  $\frac{X^2 + Y^2}{12X^2Y^2} = \frac{2(1+z)}{3\Sigma^2(1-z)^2}$ .  $\square$

**Proposition A.4.** Let  $0 < \varepsilon < \frac{1}{4}$ ,  $\kappa = \frac{2\varepsilon}{1-4\varepsilon}$ ,  $\kappa_s = \kappa(s + 10)$  and  $\Sigma_s = s + 2\kappa_s$ . For  $1 \leq s \leq n - 1$  and  $b \in \mathcal{S}_s$  let  $D_s(b) = \mathcal{L}_s \Phi_{s+1}^{\kappa_{s+1}}(b) / \Phi_s^{\kappa_s}(b)$ .

(a) Let  $c = \kappa_s$ ,  $c' = \kappa_{s+1}$ ,  $d = \frac{s}{2} - b$ ,  $u = \frac{d}{s+2c'}$ ,  $v = \frac{d}{n-s}$  and  $\rho = \frac{s+2c'}{s(1+2\beta)}$ . Then  $|u| < \frac{1}{2}$  and

$$D_s(b) = \mathcal{J} \frac{\Phi_s^{c'}(b)}{\Phi_s^c(b)}, \quad \mathcal{J} = 1 - (\rho - 1) \frac{u(u+v)}{\frac{1}{4} - u^2}.$$

(b)  $\log D_s(b) \leq \frac{\kappa}{\Sigma_s} + \frac{2\kappa}{3\Sigma_s^2}$ .

(c) For  $1 \leq t \leq h$ ,  $V_{t+1}(1) \leq \pi_t \exp(\kappa[H_{n-1} - H_t + \frac{2}{3t}] + \kappa_{t+1}[\log \frac{(t+1)^2}{4t} + \frac{2}{3}])$ .

(d)  $C_n \leq U_n := \varepsilon c_1 e^{\kappa H_{n-1}} \sum_{t=1}^h w_t (1 - \varepsilon)^{t-1} e^{\kappa \mathcal{G}(t)}$ , where  $\mathcal{G}(t) = (t+11)(\log \frac{(t+1)^2}{4t} + \frac{2}{3}) - H_t + \frac{2}{3t}$ .

The slope  $\kappa$  is chosen so that  $2(\kappa - \beta)/(1 + 2\beta) = \kappa$ . In the proof of (b) this makes the gain in  $\mathcal{J}$  cancel the main loss from raising the pseudo-count from  $c$  to  $c'$ , and the offset 10 in  $\kappa_s$  controls what is left.

*Proof.* (a) Write  $p' = p_s^{c'}(b)$ ,  $\bar{r} = \bar{r}_s(b)$  and  $r = r_s(b)$ . By Lemma A.1(a) and (b) with  $c'$ , the numerator of  $D_s(b)$  divided by  $\Phi_s^{c'}(b)$  is  $\frac{(1-r)(1-\bar{r})}{1-p'} + \frac{r\bar{r}}{p'}$ . Here  $p' = \frac{1}{2} - u$ ,  $\bar{r} = \frac{1}{2} + v$  and  $r = \frac{1}{2} - \rho u$ , since  $\frac{1}{2} - r = (1 - 2\varepsilon)\frac{d}{s} = \rho u$ . Over the denominator  $p'(1 - p') = \frac{1}{4} - u^2$  the numerator is  $f(u, v) + f(-u, -v)$  with  $f(u, v) = (\frac{1}{2} + \rho u)(\frac{1}{2} - v)(\frac{1}{2} - u)$ , and this equals  $\frac{1}{4} - \rho u^2 - (\rho - 1)uv$ . Also  $|d| \leq \frac{s}{2} - 1$ , so  $|u| < \frac{1}{2}$ .

(b) Write  $c = \kappa_s$ ,  $c' = c + \kappa$  and  $\Sigma = \Sigma_s$ . Since  $c' - \beta s = (\kappa - \beta)s + 11\kappa$ , the choice of  $\kappa$  gives  $\rho - 1 = \kappa + \nu_s$  with  $\nu_s = \frac{22\kappa}{s(1+2\beta)}$ . As  $u$  and  $v$  have the sign of  $d$ ,  $u(u+v) \geq u^2$ . So  $\mathcal{J} \leq 1 - (\kappa + \nu_s)W'$  with  $z' = 4u^2$  and  $W' = \frac{z'}{1-z'}$ . Also  $\mathcal{J} > 0$  by (a), since  $0 < r < 1$  and  $\bar{r}, 1 - \bar{r}$  are not both 0. So  $\log \mathcal{J} \leq -(\kappa + \nu_s)W'$ . Next,  $\log(\Phi_s^{c'}(b)/\Phi_s^c(b))$  is the integral over  $\tau \in [c, c']$  of the derivative in Lemma A.3. Its bound increases in  $z$  and decreases in  $\Sigma$ , while  $z$  decreases and  $\Sigma$  increases in  $\tau$ . So we may take the bound at  $\tau = c$ , where  $X + Y = \Sigma$ . Put  $W = \frac{z}{1-z}$ . Then  $\frac{1}{1-z} = 1 + W$ ,  $\frac{1+z}{1-z} = 1 + 2W$  and  $-\log(1 - z) \leq W$ , so

$$\log D_s(b) \leq \kappa \left[ W + \frac{1 + 2W}{\Sigma} + \frac{2(1 + 2W)(1 + W)}{3\Sigma^2} \right] - (\kappa + \nu_s)W'. \quad (2)$$

We use 3 facts. (i)  $X, Y \geq 1 + c$ , since  $b \geq 1$  and  $s - b \geq 1$ . So  $XY \geq (1 + c)\Sigma/2$  and  $1 + W = \frac{\Sigma^2}{4XY} \leq \frac{\Sigma}{2(1+c)}$ . (ii)  $z' = \lambda z$  with  $\lambda = (\frac{\Sigma}{\Sigma+2\kappa})^2 \geq 1 - \frac{4\kappa}{\Sigma}$ . So  $W - W' = \frac{(1-\lambda)z}{(1-z)(1-\lambda z)} \leq (1 - \lambda)W(1 + W) \leq \frac{2\kappa}{1+c}W$  by (i). As  $c \geq 11\kappa$ , this gives  $W' \geq \frac{9}{11}W$ . (iii)  $\Sigma \geq s$  and  $1 + c > \kappa s$ . If  $W > 0$  then  $s \neq 2b$ , and with  $b \geq 1$ ,  $s - b \geq 1$  this forces  $s \geq 3$ , so  $\Sigma^2 \geq 3s$ .

By (ii) and (iii),  $\kappa(W - W') \leq \frac{2\kappa^2}{1+c}W \leq \frac{2\kappa}{s}W$  and  $\frac{2\kappa W}{\Sigma} \leq \frac{2\kappa}{s}W$ . Since  $(1 + 2W)(1 + W) = 1 + 3W + 2W^2$ ,  $\kappa$  times the last term in the bracket of (2) is  $\frac{2\kappa}{3\Sigma^2} + \frac{2\kappa W}{\Sigma^2} + \frac{4\kappa W^2}{3\Sigma^2}$ . Here  $\frac{2\kappa W}{\Sigma^2} \leq \frac{2\kappa}{3s}W$  by (iii), and  $\frac{4\kappa W^2}{3\Sigma^2} \leq \frac{4\kappa W(1+W)}{3\Sigma^2} \leq \frac{2\kappa W}{3\Sigma(1+c)} \leq \frac{2\kappa}{3s}W$  by (i). With  $-\nu_s W' \leq -\frac{9}{11}\nu_s W$  this gives

$$\log D_s(b) \leq \frac{\kappa}{\Sigma} + \frac{2\kappa}{3\Sigma^2} + W \left[ \frac{16\kappa}{3s} - \frac{18\kappa}{s(1+2\beta)} \right],$$

and the bracket is negative because  $1 + 2\beta < 2$  for  $\varepsilon < \frac{1}{4}$ .

(c) Apply Lemma A.2(a) with  $s_0 = t+1$ ,  $F_s = \Phi_s^{\kappa_s}$  and  $K_s = \exp(\frac{\kappa}{s} + \frac{2\kappa}{3s^2})$ , which is allowed by (b) and  $\Sigma_s \geq s$ . Since  $1 \in \mathcal{S}_{t+1}$  and  $\sum_{s>t} s^{-2} \leq \frac{1}{t}$ , we get  $V_{t+1}(1) \leq e^{\kappa[H_{n-1}-H_t+2/(3t)]} \Phi_{t+1}^{\kappa_{t+1}}(1)$ . Now  $\log(\Phi_{t+1}^{\kappa_{t+1}}(1)/\pi_t)$  is the integral over  $\tau \in [0, \kappa_{t+1}]$  of the derivative in Lemma A.3, whose bound is largest at  $\tau = 0$ . There  $X = 1$ ,  $Y = t$  and  $\frac{1}{1-z} = \frac{(t+1)^2}{4t}$ , and the bound is  $\log \frac{(t+1)^2}{4t} + \frac{t^2+1}{2t(t+1)} + \frac{t^2+1}{12t^2} \leq \log \frac{(t+1)^2}{4t} + \frac{2}{3}$ .

(d) Insert (c) into Proposition 2.5(a), and use  $\kappa_{t+1} = \kappa(t+1)$  and  $\pi_t = c_1 w_t$ .  $\square$

*Proof of Theorem 2.6.* (a) By Proposition 2.5(a) and Lemma A.1(d),  $C_n \geq \varepsilon c_1 \sum_t w_t (1 - \varepsilon)^{t-1}$ . Now  $(1 - \varepsilon)^{t-1} \geq 1 - (t-1)\varepsilon$  and  $\sum_t w_t (t-1) = 2 - \frac{6}{h+2}$  by Proposition 2.5(c). So  $C_n \geq \varepsilon c_1 (1 - 2\varepsilon)$ , and  $c_1 = \frac{4}{n+2}$ .

(b) Let  $n \geq 64$  and  $\varepsilon(1 + \log n) \leq \frac{1}{100}$ . Then  $\varepsilon < 0.002$  and  $\kappa \leq 2\varepsilon(1 + 8\varepsilon) \leq 2.1\varepsilon$ . Since  $\log \frac{(t+1)^2}{4t} \geq 0$  and  $\frac{2}{3}(t+1) > 1 + \log t \geq H_t$ , we have  $\mathcal{G}(t) > 0$ . Since  $\frac{(t+1)^2}{4t} \leq \frac{t+1}{2}$ ,  $\frac{2}{3} < \log 2$  and  $H_t \geq \frac{2}{3t}$ , also  $\mathcal{G}(t) \leq (t+1) \log(t+1) \leq 12t \log(t+1)$ . So  $\kappa \mathcal{G}(t) \leq 25.2 \varepsilon t \log(t+1)$ , and this is at most  $\frac{t}{2} \log 2$  for  $t \leq h$ , since then  $\log(t+1) \leq \log n$ . Hence  $e^{\kappa \mathcal{G}(t)} - 1 \leq \kappa \mathcal{G}(t) 2^{t/2}$ . By Proposition A.4(d),  $(1 - \varepsilon)^{t-1} \leq 1$ ,  $\sum_t w_t = 1$  and  $w_t \leq 2t2^{-t}$  (Proposition 2.5(c)),

$$C_n \leq \varepsilon c_1 e^{\kappa H_{n-1}} \left( 1 + 50.4 \varepsilon \sum_{t \geq 1} t^2 \log(t+1) 2^{-t/2} \right),$$

and we bound the series as follows. The ratio of its consecutive terms is  $(1 + \frac{1}{t})^2 \frac{\log(t+2)}{\log(t+1)} 2^{-1/2}$ , which decreases in  $t$  and is below  $\frac{3}{4}$  at  $t = 60$ . So the terms with  $t > 60$  add up to less than 4 times the term at  $t = 61$ , which is below  $10^{-4}$ . The first 60 terms add up to 102.5864 to 4 decimals, so the series is below 102.59. Finally  $H_{n-1} \leq 1 + \log n$ , so  $\kappa H_{n-1} \leq 2\varepsilon \log n + 2\varepsilon + 16\varepsilon^2(1 + \log n) \leq 2\varepsilon \log n + 3\varepsilon$ . With  $1 + y \leq e^y$  this proves (b), since  $3 + 50.4 \cdot 102.59 < 5200$ .  $\square$

## B Proofs for Section 3

We keep the conventions of Section 3. We write  $C$  for an absolute constant whose value may change from line to line, and  $\bar{c}$ ,  $A_0$ ,  $A_1$  for fixed absolute constants. Since  $M$  is independent of  $X_1$ ,  $f_n(m) = \mathbb{P}(M_n = m)$ . Every path of a chain  $M^{[k_0]}$  is nondecreasing in time, and we use the paths of  $Y$  and  $Z^{(j)}$  fixed after Lemma 3.3. The family  $\mathcal{T}_v$  has the law  $\beta_v$  of Lemma 2.4. Let  $\bar{\beta}_v(s) = \mathbb{P}(|\mathcal{T}_v| \geq s)$ , with  $\bar{\beta}_v(s) = 1$  for  $s \leq 1$ . By Pascal's rule  $\binom{n-s}{v-1} - \binom{n-s-1}{v-1} = \binom{n-s-1}{v-2}$  the tail of  $\beta_v$  telescopes. So, for  $1 \leq s \leq n-1$ ,

$$\bar{\beta}_v(s) = \binom{n-s}{v-1} \bigg/ \binom{n-1}{v-1} = \prod_{i=1}^{v-1} \left( 1 - \frac{s-1}{n-i} \right) \leq e^{-(v-1)(s-1)/(n-1)}, \quad \beta_v(s) = \frac{v-1}{n-s} \bar{\beta}_v(s), \quad (3)$$

where the product vanishes as soon as one factor does. Where they are defined, the ratios are

$$\frac{\beta_v(d)}{\beta_v(d-1)} = 1 - \frac{v-2}{n-d}, \quad \frac{\bar{\beta}_v(s-1)}{\bar{\beta}_v(s)} = 1 + \frac{v-1}{n-s-v+2}. \quad (4)$$

**Lemma B.1** (monotone sandwich). *Let  $0 \leq a \leq b \leq n$ . If  $m_h \leq a$ , then  $(b - a + 1)f_n(b) \leq \mathbb{P}(a \leq M_n \leq b)$ . If  $b \leq m_h$ , then  $(b - a + 1)f_n(a) \leq \mathbb{P}(a \leq M_n \leq b)$ .*

*Proof.* By Lemma 3.4,  $f_n(\ell) \geq f_n(b)$  for  $a \leq \ell \leq b$  in the first case, and  $f_n(\ell) \geq f_n(a)$  in the second. Sum over  $\ell$ .  $\square$

**Lemma B.2** (tail). *Let  $0 < \varepsilon < 1$ ,  $k_0 \geq 1$  and  $N \geq k_0$ . For every integer  $z \geq 1$ ,*

$$\mathbb{P}(M_N^{[k_0]} \geq z) \leq \frac{\varepsilon N H_z}{z}, \quad \mathbb{E} M_N^{[k_0]} \leq \varepsilon N \log \frac{N}{k_0}.$$

*In particular  $\mathbb{P}(M_n \geq z) \leq x H_z / z$ , and  $\mathbb{P}(Z_t^{(j)} \geq z) \leq x H_z / z$  for  $t \leq n$ .*

*Proof.* We use the tree of Lemma 3.1 on  $[N]$ , with marks only on  $v > k_0$ . Every odd vertex lies in the family of a marked vertex, so  $M_N^{[k_0]} \leq \sum_{v \text{ marked}} |\mathcal{T}_v|$ . So  $M_N^{[k_0]} \geq z$  needs a marked family of size at least  $z$ , or  $\sum_{v \text{ marked}} |\mathcal{T}_v| \mathbf{1}\{|\mathcal{T}_v| < z\} \geq z$ . By the union bound and Markov's inequality, and since marks and tree are independent, the probability is at most  $\varepsilon \sum_v \mathbb{P}(|\mathcal{T}_v| \geq z) + \frac{\varepsilon}{z} \sum_v \mathbb{E}[|\mathcal{T}_v|; |\mathcal{T}_v| < z]$ . By the first sum in Lemma 2.4, with  $N$  in place of  $n$ , this is at most  $\varepsilon \frac{N}{z} + \frac{\varepsilon}{z} \sum_{d=1}^{z-1} \frac{N}{d+1} = \frac{\varepsilon N H_z}{z}$ . For the mean,  $\mathbb{E} M_N^{[k_0]} \leq \varepsilon \sum_{v=k_0+1}^N \frac{N}{v} \leq \varepsilon N \log \frac{N}{k_0}$ .  $\square$

**Lemma B.3** (late start). *Let  $j \geq 2$ ,  $N \geq j - 1$  and  $z \geq 0$ . Then  $\mathbb{P}(M_N^{[j-1]} \geq z) \leq \exp\{-\frac{j-1}{2}(\frac{z}{N} - \varepsilon \log \frac{2N}{j-1})\}$ .*

*Proof.* Write  $G_k = M_k^{[j-1]}$  and  $\nu = (j - 1)/2$ . For  $k > \nu$  let  $\theta_k = \log \frac{k}{k-\nu}$ , so that  $e^{\theta_k} - 1 = \frac{\nu}{k-\nu}$ . Since  $r_k(c) \leq \varepsilon + c/k$  and  $1 + \nu \leq e^\nu$ ,  $\mathbb{E}[e^{\theta_{k+1} G_{k+1}} | G_k = c] \leq \exp\{c(\theta_{k+1} + \frac{\nu}{k(k+1-\nu)}) + \frac{\varepsilon \nu}{k+1-\nu}\}$ . Now  $\theta_k - \theta_{k+1} = \log(1 + w)$  with  $w = \frac{\nu}{(k-\nu)(k+1)}$ , and  $\log(1 + w) \geq \frac{w}{1+w} = \frac{\nu}{k(k+1-\nu)}$ . So  $\mathbb{E} e^{\theta_{k+1} G_{k+1}} \leq e^{\varepsilon \nu / (k+1-\nu)} \mathbb{E} e^{\theta_k G_k}$  for  $k \geq j - 1$ . Starting from  $G_{j-1} = 0$  and using  $\sum_{k=j}^N \frac{1}{k-\nu} \leq \log \frac{N-\nu}{\nu} \leq \log \frac{2N}{j-1}$ , we get  $\mathbb{E} e^{\theta_N G_N} \leq \exp\{\varepsilon \nu \log \frac{2N}{j-1}\}$ . Markov's inequality with  $\theta_N \geq \nu/N$  gives the claim.  $\square$

**Lemma B.4** (the first flipped family). *For  $1 \leq d \leq n - 1$ ,*

$$\frac{x}{d(d+1)} \left(1 - \frac{2x(n-d-1)}{n(d+2)}\right) \leq \mathbb{P}(D = d) \leq \frac{x}{d(d+1)}, \quad \mathbb{P}(D \geq d) \leq x \left(\frac{1}{d} - \frac{1}{n}\right).$$

*Proof.* By Lemma 3.3,  $\mathbb{P}(D = d) = \sum_j \varepsilon (1 - \varepsilon)^{j-2} \beta_j(d)$ , and  $1 - (j - 2)\varepsilon \leq (1 - \varepsilon)^{j-2} \leq 1$ . Apply the 2 sums of Lemma 2.4, and sum the upper bound over  $d' \geq d$ .  $\square$

**Proposition B.5** (local lower bound). *There is an absolute constant  $C$  such that the following hold.*

- (a) *For  $2 \leq m \leq n/4$ ,  $f_n(m) \geq \frac{x}{m(m+1)}(1 - e_1(m))$  with  $e_1(m) = C\left(\frac{x \log(m+1)}{m} + \varepsilon \log n + m e^{-m/9}\right)$ .*
- (b) *For  $n/4 < m \leq n - 16$  and  $r = n - m$ ,  $f_n(m) \geq \frac{x}{m(m+1)}(1 - e_2(m))$  with  $e_2(m) = C\left(\frac{x \log n}{r} + \frac{x \log n}{m} + n e^{-r/40}\right)$ .*

*Proof.* (a) Fix  $2 \leq j \leq J := \lfloor n/8 \rfloor$  and put  $\Gamma_d = d - Y_d + Z_{n-d}^{(j)}$ . The inner sum in Lemma 3.3 is  $Q_j(m) = \mathbb{E} \sum_d \beta_j(d) \mathbf{1}\{\Gamma_d = m\}$ , and every step of  $\Gamma$  lies in  $\{-1, 0, 1\}$ . On the event  $\{Z_{n-1}^{(j)} \leq m - 1, Y_{2m} \leq m\}$  we have  $\Gamma_1 = 1 + Z_{n-1}^{(j)} \leq m$ , and at  $d^\circ = m + Y_{2m} \leq 2m$  we have  $\Gamma_{d^\circ} \geq d^\circ - Y_{2m} = m$ , since  $Y$  is nondecreasing (and  $2m + j \leq n$ ). So  $\Gamma$  hits  $m$  at some  $d \leq d^\circ$ , and as  $\beta_j$  is nonincreasing

the sum is at least  $\beta_j(m + Y_{2m})$ . By (4),  $\beta_j(m + y)/\beta_j(m) = \prod_{i=1}^y (1 - \frac{j-2}{n-m-i}) \geq 1 - \frac{2y(j-2)}{n}$  for  $y \leq m$ , since each fraction is at most  $\frac{2(j-2)}{n} \leq \frac{1}{4}$ . By independence and  $ab \geq a + b - 1$  for  $a, b \in [0, 1]$ ,  $Q_j(m) \geq \beta_j(m)[1 - \mathbb{P}(Y_{2m} > m) - \mathbb{P}(Z_{n-1}^{(j)} \geq m)] - \frac{2(j-2)}{n} \mathbb{E}Y_{2m} \beta_j(m)$ . Lemma B.2 gives  $\mathbb{P}(Y_{2m} > m) \leq 2\varepsilon H_{m+1}$ ,  $\mathbb{P}(Z_{n-1}^{(j)} \geq m) \leq xH_m/m$  and  $\mathbb{E}Y_{2m} \leq 2\varepsilon m \log n$ . If  $2\varepsilon H_{m+1} + xH_m/m > 1$ , then  $e_1(m) > 1$  and there is nothing to prove. Otherwise we multiply by  $\mathbb{P}(V = j) \leq \varepsilon$  and sum over  $2 \leq j \leq J$ . By the second sum in Lemma 2.4, the terms with  $\mathbb{E}Y_{2m}$  add up to at most  $\frac{4\varepsilon m \log n}{n} \cdot \frac{2x(n-m-1)}{m(m+1)(m+2)} \leq \frac{x}{m(m+1)} 8\varepsilon \log n$ . By Lemma B.4,  $\sum_{j \leq J} \mathbb{P}(V = j) \beta_j(m) \geq \frac{x}{m(m+1)} (1 - \frac{2x}{m+2}) - \varepsilon \sum_{j > J} \beta_j(m)$ . By (3),  $\beta_j(m) \leq \frac{j-1}{n-m} e^{-(j-1)a}$  with  $a = \frac{m-1}{n-1} \leq \frac{1}{4}$ , and  $aJ \geq \frac{m-1}{9}$  because  $\lfloor n/8 \rfloor \geq \frac{n-1}{9}$  for  $n \geq 64$ . Summing the series gives  $\varepsilon \sum_{j > J} \beta_j(m) \leq \frac{C\varepsilon}{n} (\frac{J}{a} + \frac{1}{a^2}) e^{-(m-1)/9} \leq \frac{x}{m(m+1)} Cme^{-m/9}$ . Collecting the terms gives (a).

(b) The same argument works with  $J = w = \lfloor r/8 \rfloor \geq 2$ , the event  $\{Z_{n-1}^{(j)} \leq m-1, Y_{m+w} \leq w\}$  and  $d^o = m + Y_{m+w} \leq m + w$ . Note that  $m + w + j \leq n$ . Only 3 estimates change. For  $y \leq w$  and  $j \leq J$  we have  $n - m - i \geq \frac{7}{8}r$ , so  $\beta_j(m + y) \geq \beta_j(m)(1 - \frac{8y(j-2)}{7r})$ . Since  $w + 1 \geq r/8$ , Lemma B.2 gives  $\mathbb{P}(Y_{m+w} > w) \leq 8xH_n/r$  and  $\mathbb{E}Y_{m+w} \leq x \log n$ , and the terms with  $\mathbb{E}Y_{m+w}$  add up to at most  $\frac{x}{m(m+1)} \cdot \frac{Cx \log n}{m}$ . For the dropped terms,  $a \geq \frac{1}{5}$  and  $J \geq \frac{r-7}{8}$ , so  $\varepsilon \sum_{j > J} \beta_j(m) \leq \frac{C\varepsilon(J+1)}{r} e^{-J/5} \leq \frac{x}{m(m+1)} Cne^{-r/40}$ .  $\square$

**Proposition B.6** (cumulative upper bound). *There is an absolute constant  $C$  such that for  $6 \leq m \leq n/2$  with  $m \geq 16x \log m$  we have  $\mathbb{P}(M_n \geq m) \leq \frac{x}{m} (1 + C \frac{x \log^2 m}{m})$ .*

*Proof.* Let  $J = \lfloor n/4 \rfloor$  and  $Z_j = Z_{n-1}^{(j)}$ . On  $\{V = j\}$  we have  $M_n = D - Y_D + Z_{n-D}^{(j)} \leq D + Z_j$ , and  $D$  is independent of  $Z_j$  with law  $\beta_j$ . On  $\{V > J\}$  there is no mark on  $2, \dots, J$ , so  $M_n$  has the law of  $M_n^{[J]}$  by Lemma 3.1. For  $0 \leq z \leq m/8$ , (4) gives  $\bar{\beta}_j(m - z) \leq \bar{\beta}_j(m) \varrho_j^z$  with  $\varrho_j = 1 + \frac{j-1}{n-m-j+2} \leq 1 + \frac{4(j-1)}{n}$ , and  $\varrho^z - 1 \leq z(\varrho - 1)\varrho^z$ . Hence

$$\mathbb{P}(M_n \geq m) \leq \sum_{j=2}^J \mathbb{P}(V = j) \left[ \bar{\beta}_j(m) (1 + (\varrho_j - 1) \varrho_j^{m/8} \mathbb{E}Z_j) + \mathbb{P}\left(Z_j > \frac{m}{8}\right) \right] + \mathbb{P}(V > J) \mathbb{P}(M_n^{[J]} \geq m).$$

This gives 4 terms. The main term is  $\sum_j \mathbb{P}(V = j) \bar{\beta}_j(m) \leq \mathbb{P}(D \geq m) \leq \frac{x}{m}$  by Lemma B.4.

For the shift term we use  $\mathbb{P}(V = j) \leq \varepsilon$ ,  $\bar{\beta}_j(m) \leq e^{-(j-1)(m-1)/(n-1)}$ ,  $\varrho_j^{m/8} \leq e^{(j-1)m/(2n)}$  and  $\mathbb{E}Z_j \leq x \log \frac{n}{j-1}$  (Lemma B.2). As  $m \geq 6$ , the exponents add up to at most  $-(j-1)a$  with  $a = \frac{m}{3n}$ . With  $i = j-1$  and  $\log \frac{n}{i} = \log \frac{m}{3} + \log \frac{1}{ai}$ , the shift term is at most  $\frac{4\varepsilon x}{n} [\log \frac{m}{3} \sum_{i \geq 1} i e^{-ai} + \sum_{i \leq 1/a} i \log \frac{1}{ai}] \leq \frac{4\varepsilon x}{n} \cdot \frac{\log m + C}{a^2} \leq \frac{x}{m} \cdot \frac{Cx \log m}{m}$ , since  $\sum_i i e^{-ai} \leq a^{-2}$  and  $t \log \frac{1}{at} \leq \frac{1}{ea}$  for  $0 < t \leq 1/a$ .

For the big- $Z$  term let  $\mathcal{K} = 32 \log(m/x)$  and  $j_{\mathcal{K}} = \lfloor \mathcal{K}n/m \rfloor + 1$ . Note that  $m \geq 16x \log 6 > 28x$ , so  $\mathcal{K} \geq 2$ . For  $j \leq j_{\mathcal{K}}$ , Lemma B.2 gives  $\mathbb{P}(Z_j > m/8) \leq 8xH_m/m$ , and these  $j$  contribute at most  $\varepsilon \frac{\mathcal{K}n}{m} \cdot \frac{8xH_m}{m} \leq \frac{x}{m} \cdot \frac{Cx \log^2 m}{m}$ . For  $j > j_{\mathcal{K}}$  we have  $\frac{2(n-1)}{j-1} \leq \frac{2m}{\mathcal{K}} \leq m$ , so  $\varepsilon \log \frac{2(n-1)}{j-1} \leq \frac{x \log m}{n} \leq \frac{m}{16n}$ , and Lemma B.3 with  $N = n-1$  gives  $\mathbb{P}(Z_j \geq m/8) \leq e^{-(j-1)m/(32n)}$ . Summing over  $j-1 \geq j_{\mathcal{K}} > \mathcal{K}n/m$  gives at most  $\varepsilon \frac{e^{-\mathcal{K}/32}}{1 - e^{-m/(32n)}} \leq \varepsilon \left( \frac{32n}{m} + 1 \right) \frac{x}{m} \leq \frac{x}{m} \cdot \frac{33x}{m}$ .

Finally  $\mathbb{P}(V > J) = (1 - \varepsilon)^{J-1} \leq e^{-3x/16}$ , since  $J \geq 3n/16$ , and Lemma B.3 with  $j-1 = J \in [\frac{3n}{16}, \frac{n}{4}]$  and  $N = n$  gives  $\mathbb{P}(M_n^{[J]} \geq m) \leq \exp\{-\frac{3m}{32} + \frac{x}{8} \log \frac{32}{3}\}$ . As  $x \leq m/28$ , the last term is at most  $e^{1/4 - m/13} \leq \frac{x}{m} \cdot \frac{Cx \log^2 m}{m}$ .  $\square$

**Lemma B.7** (mode bound). *There are absolute constants  $\bar{c} > 0$  and  $A_0$  such that if  $\varepsilon \log n \leq \bar{c}$  and  $A_0 x \log(x+2) \leq n/4$ , then  $m_h \leq A_0 x \log^2(x+1)$ .*

*Proof.* Let  $\ell = \lceil Ax \log(x+2) \rceil$ . Since  $\log(\ell+1) \leq \log(A+2) + 2\log(x+2)$ , we have  $\frac{x \log(\ell+1)}{\ell} \leq \frac{\log(A+2)+2}{A}$ , and  $\ell e^{-\ell/9}$  is small when  $A$  is large. So we can fix the absolute constants  $A$  and  $\bar{c}$  such that  $e_1(\ell) \leq \frac{1}{2}$  in Proposition B.5(a) whenever  $\varepsilon \log n \leq \bar{c}$ . We take  $A_0 \geq A+1$ , so that  $2 \leq \ell \leq n/4$ . Then  $f_n(\ell) \geq \frac{x}{2\ell(\ell+1)}$ . If  $m_h \geq \ell$ , Lemmas B.1 (with  $a = \ell$ ,  $b = m_h$ ) and B.2 give  $(m_h - \ell + 1)f_n(\ell) \leq \mathbb{P}(M_n \geq \ell) \leq xH_\ell/\ell$ . So  $m_h \leq (\ell+1)(1+2H_\ell)$  in every case. As  $\log \ell \leq C \log(x+2)$  and  $\log(x+2) \leq 2\log(x+1)$ , this is at most  $A_0 x \log^2(x+1)$  once  $A_0$  is large.  $\square$

**Proposition B.8** (local upper bound). *There are absolute constants  $A_1$  and  $C$  such that if  $\varepsilon \log n \leq \bar{c}$  and  $A_1 x \log^2(x+1) \leq m \leq n/8$ , then*

$$f_n(m) \leq \frac{x}{m(m+1)} (1 + C\Delta(m)^{1/2}), \quad \Delta(m) = \frac{x \log^2 m}{m} + \varepsilon \log n + \frac{m}{n}.$$

*Proof.* We take  $A_1 \geq 2A_0$  so large that Lemma B.7 applies and every  $m' \geq m/2$  satisfies  $m' \geq 6$  and  $m' \geq 16x \log m'$  (as  $t/\log t$  increases for  $t \geq e$ ). Let  $1 \leq k \leq m/2$ . Then  $m-k \geq m/2 \geq m_h$ , and Lemma B.1 with  $a = m-k$  and  $b = m$  gives  $(k+1)f_n(m) \leq \mathbb{P}(M_n \geq m-k) - \mathbb{P}(M_n \geq m+1)$ . Proposition B.6 at  $m-k$  bounds the first term by  $\frac{x}{m-k} + \frac{Cx^2 \log^2 m}{m^2}$ . For the second we sum Proposition B.5(a) over  $m < \ell \leq n/4$ . Since  $\sum_{m < \ell \leq n/4} \frac{1}{\ell(\ell+1)} \geq \frac{1}{m+1} - \frac{4}{n}$ ,  $\frac{\log(\ell+1)}{\ell}$  and  $\ell e^{-\ell/9}$  decrease for  $\ell \geq 9$ , and  $me^{-m/9} \leq C/m \leq C\Delta(m)$ , we get  $\mathbb{P}(M_n \geq m+1) \geq \frac{x}{m+1}(1 - C\Delta(m))$ . As  $\frac{x}{m-k} - \frac{x}{m+1} = \frac{x(k+1)}{(m-k)(m+1)}$ , this gives  $f_n(m) \leq \frac{x}{(m-k)(m+1)} + \frac{Cx\Delta(m)}{(k+1)m}$ . If  $\Delta(m) \leq \frac{1}{16}$ , take  $k = \lceil m\Delta(m)^{1/2} \rceil \leq \frac{m}{2}$ . Then  $\frac{m}{m-k} \leq 1 + \frac{2k}{m} \leq 1 + 2\Delta^{1/2} + \frac{2}{m}$  and  $\frac{\Delta}{k+1} \leq \frac{\Delta^{1/2}}{m}$ , which gives the claim, as  $\frac{1}{m} \leq \Delta(m) \leq \Delta(m)^{1/2}$ . If  $\Delta(m) > \frac{1}{16}$ , take  $k = \lfloor m/2 \rfloor$ . Then  $f_n(m) \leq \frac{x}{m(m+1)}(2 + C\Delta(m))$ , and the claim follows after enlarging  $C$ , since  $\Delta(m)$  is bounded by an absolute constant ( $\frac{x \log^2 m}{m}$  decreases for  $m \geq e^2$  and is bounded at  $m = A_1 x \log^2(x+1)$ ).  $\square$

*Proof of Theorem 3.8.* Put  $\ell_x = \log(x+1) \geq \log 2$  and let  $\mathcal{R}_n$  be the range  $\omega_n x \ell_x^2 \leq m \leq \delta'_n n$ . Since  $x \log^2 n/n \rightarrow 0$ , for large  $n$  we have  $\varepsilon(1 + \log n) \leq \min(\bar{c}, \frac{1}{100})$ ,  $\omega_n \geq \max(A_1, 16)$ ,  $\delta'_n \leq \frac{1}{8}$  and  $A_0 x \log(x+2) \leq n/4$ .

The function  $\frac{x \log^2 m}{m}$  decreases for  $m \geq e^2$ . At  $m = \omega_n x \ell_x^2$  we have  $m \geq e^2$  and  $\log m \leq \log \omega_n + 3\ell_x$ , as  $\log x$  and  $\log \ell_x$  are at most  $\ell_x$ . So on  $\mathcal{R}_n$ , uniformly in  $x \geq 1$ ,  $\frac{x \log^2 m}{m} \leq \frac{(\log \omega_n + 3\ell_x)^2}{\omega_n \ell_x^2} \leq \frac{1}{\omega_n} (\frac{\log \omega_n}{\log 2} + 3)^2 \rightarrow 0$ . Also  $m \geq \omega_n \log^2 2 \rightarrow \infty$  on  $\mathcal{R}_n$ . Since  $\log(m+1) \leq 2\log^2 m$  for  $m \geq 3$ , Proposition B.5(a) has  $e_1(m) \leq C(\frac{x \log^2 m}{m} + \varepsilon \log n + me^{-m/9}) \rightarrow 0$  uniformly on  $\mathcal{R}_n$ . Also  $\Delta(m) \leq \frac{x \log^2 m}{m} + \varepsilon \log n + \delta'_n \rightarrow 0$  there, and Proposition B.8 applies. So  $f_n(m) = \frac{x}{m(m+1)}(1 + o(1))$  uniformly on  $\mathcal{R}_n$ .

For the mirror term, Lemma B.7 gives  $m_h \leq A_0 x \ell_x^2 \leq h$  for large  $n$ , since  $x \ell_x^2 \leq x \log^2(n+1) = o(n)$ . As  $n-m \geq h$ , Lemma 3.4 gives  $0 \leq f_n(n-m) \leq f_n(h) = C_n$ . By Theorem 2.6(b) and  $\varepsilon \log n \rightarrow 0$ ,  $C_n \leq \frac{4x}{n(n+2)} n^{2\varepsilon} e^{5200\varepsilon} = \frac{4x}{n^2}(1 + o(1))$ . On  $\mathcal{R}_n$  this is at most  $\frac{x}{m(m+1)} \cdot 8\delta_n'^2(1 + o(1))$ , since  $m(m+1) \leq 2\delta_n'^2 n^2$ . Hence  $\mathbb{P}(A_n = n-m) = \frac{1}{2}[f_n(m) + f_n(n-m)] = \frac{x}{2m(m+1)}(1 + o(1))$  uniformly on  $\mathcal{R}_n$ , and  $m(m+1) = m^2(1 + o(1))$  because  $m \rightarrow \infty$ .  $\square$

## C Proof of Theorem 4.4

We use the notation of Section 4, with  $1 < \alpha < 2$ ,  $a_m = m^{1/\alpha}$  and  $\tau_0 = 0$ . For a first run  $D$  let  $\tau_m^D = D + \tau'_{m-1}$ , where  $(\tau'_m)$  is an independent copy of  $(\tau_m)$ , and  $\tau_0^D = 0$ . Then  $u_m^D(k) = \mathbb{P}(\tau_m^D = k)$

and  $q_m^D(k) = \mathbb{P}(\tau_{m-1}^D < k \leq \tau_m^D)$ . Conditioning on the last run gives, for  $m, k \geq 1$ ,

$$\begin{aligned} q_m(k) &= \sum_{j \geq 1} \bar{F}(j) u_{m-1}(k-j), & q_{m+1}^D(k) &= \sum_{j \geq 1} \bar{F}(j) u_m^D(k-j), \\ u_m^D(k) &= \sum_{d \geq 1} \mathbb{P}(D=d) u_{m-1}(k-d). \end{aligned} \tag{5}$$

Since  $\tau_m$  and  $\tau_m^D$  increase strictly,  $\sum_m q_m(k) = \sum_m q_m^D(k) = 1$ ,  $\sum_m u_m^D(k) \leq 1$  and  $\sum_{m \leq m'} q_m(k) = \mathbb{P}(\tau_{m'} \geq k)$ . We put  $z_{m,k} = (k - m\mu)/a_m$  and  $\delta_m = \sup_k |a_m u_m(k) - g(z_{m,k})|$ , so that  $\delta_m \rightarrow 0$  by Lemma 4.3(iii). Let  $\omega$  be the modulus of continuity of  $g$  and  $\bar{g}(x) = \sup_{|y| \geq x} g(y)$ . By Lemma 4.3(ii),  $\omega(\eta) \rightarrow 0$  as  $\eta \rightarrow 0$  and  $\bar{g}(x) \rightarrow 0$  as  $x \rightarrow \infty$ .

**Lemma C.1.** *There is a constant  $C_B = C_B(\alpha)$  such that  $u_m(k) \leq C_B m k^{-\alpha-1}$  for all  $m \geq 1$  and  $k \geq 2m\mu$ .*

*Proof.* Here  $C$  depends only on  $\alpha$  and may change from line to line. Let  $r = 4 + 4/(\alpha - 1)$  and  $y = k/r$ .

*One big jump.* By the mean value theorem  $p_\ell \leq \alpha \ell^{-\alpha-1}$ . So for each  $i$  the probability  $\mathbb{P}(\tau_m = k, \xi_i \geq y) = \sum_{\ell \geq y} p_\ell \mathbb{P}(\tau_m - \xi_i = k - \ell)$  is at most  $\alpha r^{\alpha+1} k^{-\alpha-1}$ , and the  $m$  indices give at most  $\alpha r^{\alpha+1} m k^{-\alpha-1}$ .

*No big jump.* We may take  $y \geq 1$ , since otherwise  $\max_i \xi_i < y$  is impossible. Let  $x_0 > 0$  and  $\lambda = s/y$  with  $s \geq 0$ . Markov's inequality and  $e^x \leq 1 + x + \frac{x^2}{2} e^x$  for  $x \geq 0$  give

$$\mathbb{P}(\tau_m \geq x_0, \max_{i \leq m} \xi_i < y) \leq \exp\left(-\frac{s x_0}{y} + m \left( \frac{s}{y} \mathbb{E}[\xi; \xi < y] + \frac{s^2 e^s}{2y^2} \mathbb{E}[\xi^2; \xi < y] \right)\right). \tag{6}$$

We take  $x_0 = k$  and use  $\mathbb{E}[\xi; \xi < y] \leq \mu$ ,  $\mathbb{E}[\xi^2; \xi < y] \leq \alpha \sum_{\ell < y} \ell^{1-\alpha} \leq C y^{2-\alpha}$  and  $k - m\mu \geq k/2$ . The exponent is then at most  $-sr/2 + Cr^\alpha s^2 e^s m/k^\alpha$ . With  $s = \frac{1}{2} \log(k^\alpha/m)$ , which is positive since  $m \leq k/2 < k^\alpha$ , we have  $e^s m/k^\alpha = e^{-s}$  and  $s^2 e^{-s} \leq 1$ . So the bound is  $C(m/k^\alpha)^{r/4}$ . As  $r/4 = 1 + 1/(\alpha - 1)$  and  $m \leq k$ , this is at most  $C(m/k^\alpha)(k^{1-\alpha})^{1/(\alpha-1)} = C m k^{-\alpha-1}$ .  $\square$

**Lemma C.2.** *Let  $D$  be any a.s. finite first run. Then*

$$\sup_k \left| \frac{a_m}{\mu} q_m(k) - g(z_{m,k}) \right|, \quad \sup_k |a_m u_m^D(k) - g(z_{m,k})|, \quad \sup_k \left| \frac{a_m}{\mu} q_{m+1}^D(k) - g(z_{m,k}) \right|$$

*tend to 0 as  $m \rightarrow \infty$ . For  $m \geq 2$ ,  $\sup_k q_m(k) \leq \mu \bar{u}/a_{m-1}$  and  $\sup_k u_m^D(k) \leq \bar{u}/a_{m-1}$ .*

*Proof.* By (5),  $q_m$  and  $u_m^D$  are mixtures of shifts of  $u_{m-1}$  with total weights  $\mu$  and 1. This gives the bounds for  $m \geq 2$ .

Fix  $J \geq 1$  and  $Z > 0$ , put  $\sigma_J = \sum_{j > J} \bar{F}(j)$  and write  $z = z_{m,k}$ . In the sum for  $q_m(k)$  the terms with  $j > J$  add up to at most  $\bar{u} \sigma_J / a_{m-1}$ . For  $j \leq J$  we have  $|a_{m-1} u_{m-1}(k-j) - g(z_{m-1,k-j})| \leq \delta_{m-1}$  and  $z_{m-1,k-j} = z a_m / a_{m-1} + (\mu - j) / a_{m-1}$ . If  $|z| \leq Z$ , then  $|z_{m-1,k-j} - z| \leq \eta_m = Z(a_m/a_{m-1} - 1) + (J + \mu)/a_{m-1}$ . If  $|z| > Z$  and  $(J + \mu)/a_{m-1} \leq Z/2$ , then  $g(z)$  and  $g(z_{m-1,k-j})$  both lie in  $[0, \bar{g}(Z/2)]$ . Since  $\mu - \sum_{j \leq J} \bar{F}(j) = \sigma_J$ , we get, uniformly in  $k$  for large  $m$ ,

$$|a_{m-1} q_m(k) - \mu g(z)| \leq \mu (\delta_{m-1} + \omega(\eta_m) + \bar{g}(Z/2)) + (\bar{u} + \|g\|_\infty) \sigma_J.$$

We let  $m \rightarrow \infty$ , then  $J, Z \rightarrow \infty$ , and use  $a_m/a_{m-1} \rightarrow 1$ . This gives the first limit. The second follows in the same way from the sum for  $u_m^D$ , with the weights  $\mathbb{P}(D=d)$  and with  $\mathbb{P}(D > J)$  in place of  $\sigma_J$ . The third follows from the sum for  $q_{m+1}^D$ , with  $a_m u_m^D$  in place of  $a_{m-1} u_{m-1}$ , the second limit in place of  $\delta_{m-1} \rightarrow 0$ , and the shift  $z_{m,k-j} - z_{m,k} = -j/a_m$ .  $\square$

*Proof of Theorem 4.4.* Put  $b_h = (h/\mu)^{1/\alpha}$ . By Lemma 4.3(iv) it suffices to show that  $C_n \sim 2b_h^{-1} \int g^2$ . Note that  $b_h = o(h)$ , that  $a_m \geq b_h/8$  for  $m \geq h/(8\mu)$  and  $a_m \geq b_h/2$  for  $m \geq h/(2\mu)$ , and that  $a_m/b_h \rightarrow 1$  uniformly in  $m$  with  $|m\mu - h| \leq 2Mb_h$ , for each fixed  $M$ . By Proposition 4.2,  $C_n = \sum_{m \geq 1} [u_m(h)q_{m+1}^D(h) + u_m^D(h)q_m(h)]$ . Fix  $M > 1$ , put  $m_{\pm} = (h \pm Mb_h)/\mu$ , and let  $\Sigma_1, \dots, \Sigma_4$  be the parts of this sum over the regions  $\mathcal{R}_1 = \{m \leq h/(4\mu)\}$ ,  $\mathcal{R}_2 = \{h/(4\mu) < m < m_-\}$ ,  $\mathcal{R}_3 = \{m_- \leq m \leq m_+\}$  and  $\mathcal{R}_4 = \{m > m_+\}$ .

*Region  $\mathcal{R}_1$ .* Here  $h \geq 4m\mu$ , so  $u_m(h) \leq C_B m h^{-\alpha-1} \leq C_B h^{-\alpha}/(4\mu)$  by Lemma C.1. Also  $q_1(h) = h^{-\alpha}$ . For  $m \geq 2$  we split the sum for  $q_m(h)$  at  $j = h/2$ . The terms with  $j \geq h/2$  add up to at most  $(h/2)^{-\alpha}$ . For  $j < h/2$  we have  $h - j > h/2 \geq 2\mu(m-1)$ , so the other terms add up to at most  $\mu C_B (m-1)(h/2)^{-\alpha-1} \leq 2^{\alpha-1} C_B h^{-\alpha}$ . Since  $\sum_m q_{m+1}^D(h) \leq 1$  and  $\sum_m u_m^D(h) \leq 1$ , we get  $\Sigma_1 = O(h^{-\alpha})$ . This is  $o(1/b_h)$ , since  $1/\alpha < 1 < \alpha$ .

*Region  $\mathcal{R}_2$ .* For large  $h$  we have  $m-1 \geq h/(8\mu)$  here, so  $u_m(h), u_m^D(h) \leq \bar{u}/a_{m-1} \leq 8\bar{u}/b_h$  by Lemmas 4.3(iii) and C.2. Let  $m' = \lfloor m_- \rfloor$ . Then  $\sum_{m \in \mathcal{R}_2} q_m(h) \leq \mathbb{P}(\tau_{m'} \geq h)$  and  $\sum_{m \in \mathcal{R}_2} q_{m+1}^D(h) \leq \mathbb{P}(D + \tau_{m'} \geq h)$ , which is at most  $\mathbb{P}(D \geq b_h) + \mathbb{P}(\tau_{m'} \geq h - b_h)$ . Since  $m'\mu \leq h - Mb_h$  and  $a_{m'} \leq b_h$ , the event  $\{\tau_{m'} \geq h - b_h\}$  implies  $(\tau_{m'} - m'\mu)/a_{m'} \geq M - 1$ . As  $D$  is a.s. finite and  $(\tau_{m'} - m'\mu)/a_{m'} \Rightarrow G$ , we get  $\limsup_h b_h \Sigma_2 \leq 16\bar{u} \mathbb{P}(G \geq M - 1)$ .

*Region  $\mathcal{R}_4$ .* For large  $h$  we have  $m-1 \geq h/(2\mu)$  here, so  $u_m(h), u_m^D(h) \leq 2\bar{u}/b_h$ . Let  $k = \lfloor m_+ \rfloor$ . Then  $\sum_{m \in \mathcal{R}_4} q_m(h) = \mathbb{P}(\tau_k < h)$  and  $\sum_{m \in \mathcal{R}_4} q_{m+1}^D(h) = \mathbb{P}(D + \tau_k' < h) \leq \mathbb{P}(\tau_k < h)$ . Now  $(h - k\mu)/a_k \leq (\mu - Mb_h)/a_k \rightarrow -M$ , and the law of  $G$  is continuous, so  $\limsup_h b_h \Sigma_4 \leq 4\bar{u} \mathbb{P}(G \leq -M)$ .

*Region  $\mathcal{R}_3$ .* Here  $m \rightarrow \infty$  and  $a_m = b_h(1 + o(1))$  uniformly. By Lemma C.2 and  $\delta_m \rightarrow 0$ , uniformly in  $m \in \mathcal{R}_3$ ,

$$u_m(h)q_{m+1}^D(h) + u_m^D(h)q_m(h) = \frac{2\mu}{a_m^2} (g(z_{m,h})^2 + o(1)).$$

Put  $y_m = (h - m\mu)/b_h$ . Then  $|z_{m,h} - y_m| \leq M|b_h/a_m - 1| \rightarrow 0$ , so  $g(z_{m,h})^2 = g(y_m)^2 + o(1)$ . The  $y_m$  with  $m \in \mathcal{R}_3$  are the points of an arithmetic progression with step  $\mu/b_h$  that lie in  $[-M, M]$ . Hence

$$b_h \Sigma_3 = 2(1 + o(1)) \frac{\mu}{b_h} \sum_{m \in \mathcal{R}_3} (g(y_m)^2 + o(1)) \rightarrow 2 \int_{-M}^M g^2.$$

Adding the 4 regions,

$$\limsup_{h \rightarrow \infty} \left| b_h C_n - 2 \int g^2 \right| \leq 2 \int_{|z| > M} g^2 + 16\bar{u} (\mathbb{P}(G \geq M - 1) + \mathbb{P}(G \leq -M)),$$

and letting  $M \rightarrow \infty$  gives  $C_n \sim 2b_h^{-1} \int g^2$ .  $\square$

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